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Oracle Discoverer Desktop Tutorial

Oracle Discoverer Desktop is a data access tool that enables business professionals to view information in their company database. Discoverer is easy to use and provides a logical, intuitive interface for data query, analysis and reporting.

This tutorial walks you through the basic functions of Discoverer. Each step assumes that you have completed the previous step before proceeding.

The topics covered in this tutorial include:

- Using the Sample Workbook
- Opening a Workbook
- Formatting Data Display
- Drilling into Data
- Pivoting Data for Comparison and Analysis
- Graphing Data Trends
- Printing and Saving your Customized Views
- Exposing Exceptions in your Data
- Drilling to Related Data
- Renaming a Worksheet
- Exporting your Data to other Formats
- Creating a Workbook
- Performing Calculations
- Scheduling a Workbook

This tutorial covers the basic steps to get you up and running with Discoverer Desktop. For information on additional features not covered here, see the Oracle Discoverer Desktop User's Guide or the online Help system.

1.1 Using the Sample Workbook

You are working at the headquarters of a video rental and sales company. You wish to investigate trends in profits arising from video rentals and video sales over the past three Years. You will query information from a database that has transactional information for the 36 months in 1998, 1999, and 2000 from your 20 stores nationwide.

The sample Discoverer Workbook, **Video Stores Analysis** (Vidstr5.dis), contains information about sales for your company. You will use the **Video Stores Analysis** workbook to analyze trends in profits arising from video rentals and sales over the past three years from 20 stores nationwide. You will compare the sales between different Regions, calculate totals and percentages to analyze your sales data, and format your results for easy and attractive viewing.

1.2 Opening a Workbook

Think of a Workbook as a three-ring binder of pages each with a specific set of data for your business analysis. Your Discoverer manager has already created a Workbook containing the information that you need. All you have to do is open the Workbook.

Note: To create a new Workbook to meet your individual requirements, see Section 1.13,

"Creating a Workbook".

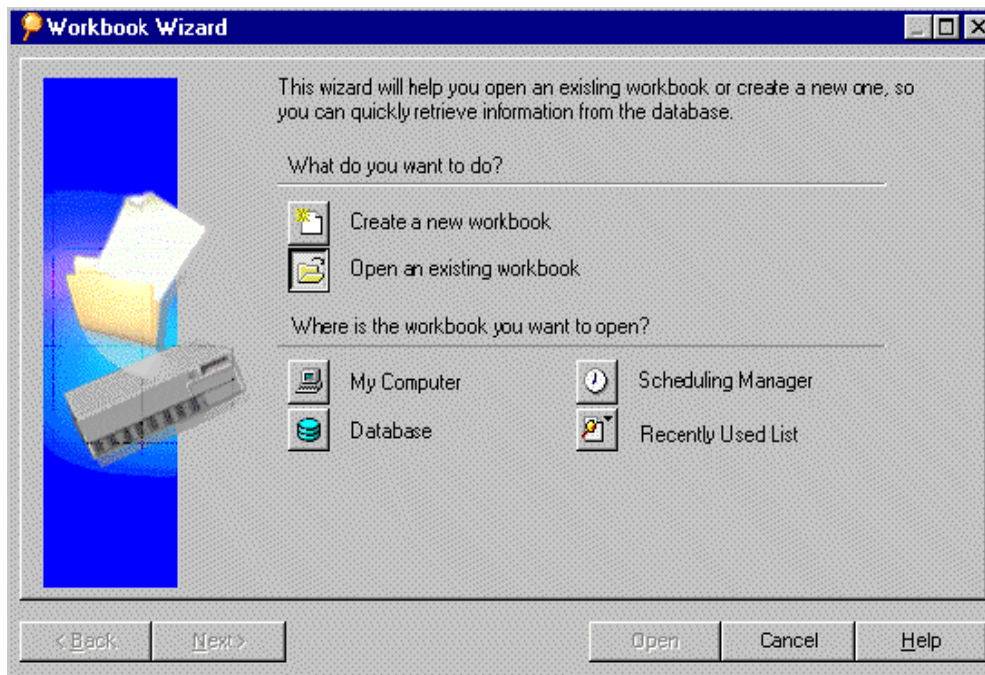
1.2.1 To start the tutorial and open a Workbook

1. From the Windows Start menu, choose **Programs | Oracle Developer Suite - <HOME_NAME> | Discoverer Desktop | Discoverer Desktop**.



2. In the Connect to Oracle Discoverer dialog box, enter your Username, Password, and the name of the Database containing the information you want. Ask your Discoverer manager for this information.

3. Click **Connect**. The Workbook Wizard appears.
In this wizard you open existing Workbooks or create new ones.



4. Select **Open an existing Workbook**.

5. Select **My Computer**. Discoverer displays a file Open dialog.
6. Navigate to the file [ORACLE_HOME]\discoverer\demo\Vidstr5.dis, and click **Open**.
[ORACLE_HOME] is the directory for your Oracle products set up by your Discoverer manager, (typically named \orant\).
7. If a **Workbook in Other Database Account** dialog is displayed, click **OK** to accept the default setting **Open the Workbook in the current database account**.
8. If you are asked whether you wish to run this query, click **Yes**.
The Video Store Workbook opens. This is a tabular report of sales information for your video sales and rental business. You are looking at the performance of the video rental and video sales Departments for the Year 1998. You can quickly see that the sales performance of the East Region is better than that of other Regions. You also see that the total profit across all three Regions for 1998 is \$391,104.
9. Click **File | Save As**, click the **My Computer** icon and then click **Save....**
10. Save this file as **Video Stores Analysis.dis**.

Note: You can also save your Workbook to the Database, in which case the *.DIS file extension is not required.

1.2.2 To Change the View using Page Items

The Worksheet you just opened shows the profits for 1998, (defined by the Page Item **Year:1998** at the top of the sheet). You want to see the profits for 2000.

1. Click on the down arrow of the Page Item **Calendar Year 1998**.

Oracle Discoverer - [Video Stores Analysis]

File Edit View Sheet Format Tools Graph Window Help

Analysis of Video Rentals and Video Sales

Page Items: Year: 1998

	Region	Department	Profit SUM
1	Central	Video Rental	\$47,204
2		Video Sale	\$67,084
3			Total for Central: \$114,288
4	East	Video Rental	\$71,766
5		Video Sale	\$108,558
6			Total for East: \$180,324
7	West	Video Rental	\$39,395
8		Video Sale	\$57,096
9			Total for West: \$96,491
10			Total for All Values: \$391,104

Tabular Layout Crosstab Layout

NUM

2. Select **2000** from the pull down list.

Page Items: Year: 1998

	Region	Department	Profit SUM
1	Central	Video Rental	\$47,204
2		Video Sale	\$67,084
3			Total for Central: \$114,288

Now you see the regional profits for the Year 2000. Notice that the profits for each Region decreased from the Year 1998, with the total of all Regions being \$320,301.

Page Items: Year: 2000 ▼			
	Region	Department	Profit SUM
▶ 1	Central	Video Rental	\$25,157
▶ 2		Video Sale	\$69,493
3			Total for Central: \$94,651
▶ 4	East	Video Rental	\$40,402
▶ 5		Video Sale	\$109,637
6			Total for East: \$150,038
▶ 7	West	Video Rental	\$23,521
▶ 8		Video Sale	\$52,092
9			Total for West: \$75,613
10			Total for All Values: \$320,301

1.3 Formatting Data Display

You can personalize the way your data is displayed by changing fonts, colors, and sizes. You can also add or edit the title of your sheet, change heading colors, format the text of your data, or add lines and spaces to view groups of data more easily.

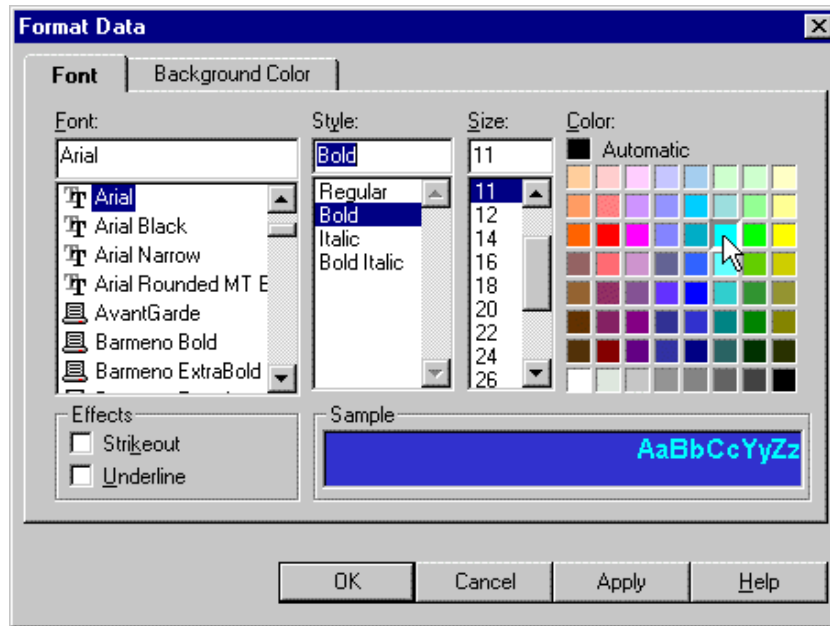
1.3.1 To Edit Data Fonts

In this section you will make your report easier to view and more attractive by editing the format of your data and your title.

1. In the **Video Stores Analysis** Workbook, click the Tabular Layout and choose the Year **1998** from the Page Items area at the top of the Worksheet.
2. Select the cell labeled **Total for Central: \$114,288**, in the Profit SUM column. The cursor becomes a white cross.

Page Items: Year: 1998 ▼			
	Region	Department	Profit SUM
▶ 1	Central	Video Rental	\$47,204
▶ 2		Video Sale	\$67,084
3			Total for Central: \$114,288

3. Choose **Format | Data**. The Format Data dialog box appears with the Font tab automatically selected.



4. Change **Size** to **11**.

Note: You can click **Apply** to make changes without closing the dialog box.

5. Change the font color to a light blue.

Look at the Sample panel to see how the text will look on your Worksheet.

6. Click **OK**.

You have now emphasized the Totals data points to make your Worksheet more readable.

Page Items: Year: 1998 ▼			
	Region	Department	Profit SUM
1	Central	Video Rental	\$47,204
2		Video Sale	\$67,084
3			Total for Central: \$114,288
4	East	Video Rental	\$71,766
5		Video Sale	\$108,558
6			Total for East: \$180,324
7	West	Video Rental	\$39,395
8		Video Sale	\$57,096
9			Total for West: \$96,491
10			Total for All Values: \$391,104

You can change any of data and heading formatting options at any time. Choose from Discoverer's wide range of fonts, colors, effects, and alignment styles.

6. Click **Edit | Undo** to undo the changes that you have made to the Workbook, or close the

Workbook without saving the changes.

1.3.2 To Edit the Title of your sheet

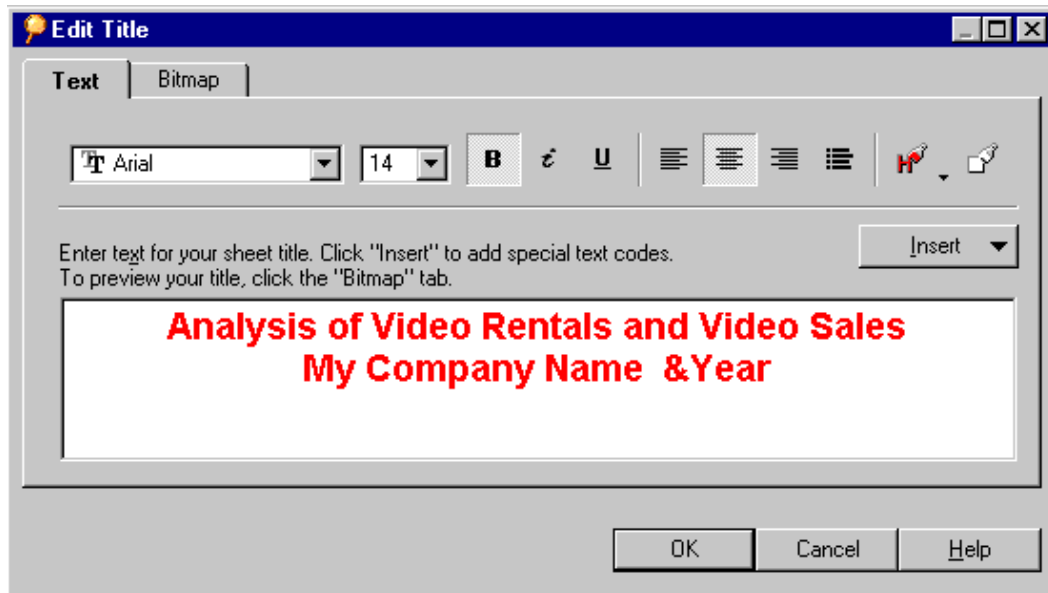
The present title of Vidstr5.dis is neutral. You can personalize the report by adding your company's title, as well as variables and graphics.

1. Choose **Sheet | Edit Title**. The Edit Title dialog box opens.
2. Add another line of text to the title by placing the cursor after the title and pressing Return. Type your video company's name-be creative!
3. Highlight the title text, "Analysis of Video Rentals ad Video Sales." Change the size to 14 pt and the color of the font to red.

Tip: Use the **Font Color Button** at the top right of the dialog box to change the font color. The color palette appears for you to select the color you want.

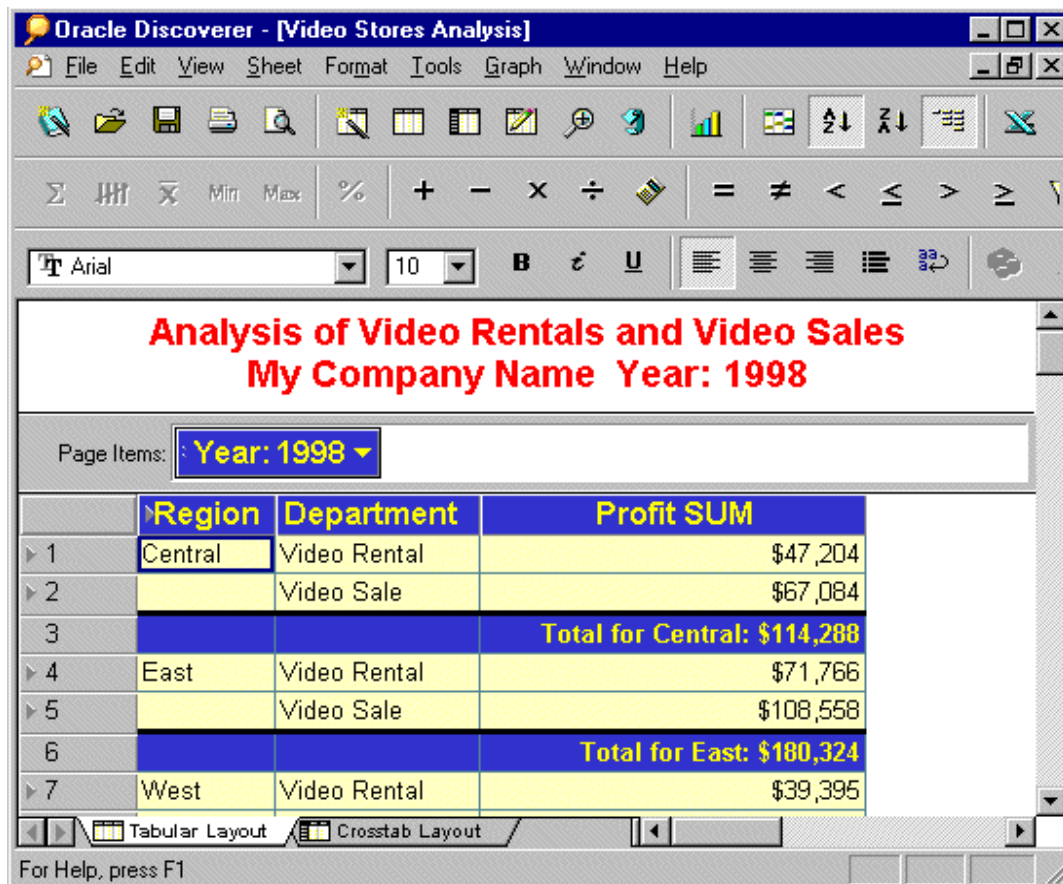
1.3.3 To Insert a Variables into your Title

1. While still in the Text page of the Edit Title dialog box, place your cursor at the end of your company's name and press the space bar to insert a space.
2. Click the **Insert** pull down list, and select **Page Items. &Year** appears next to your company name.
The variable "&Year" is dynamic, and changes according to which Year you are looking at on the Worksheet. That is, if you are looking at data for the Year 2000, '2000' will appear in your title. If you are looking at data for the Year 1999, '1999' will appear in your title and so on.



3. Click **OK**.

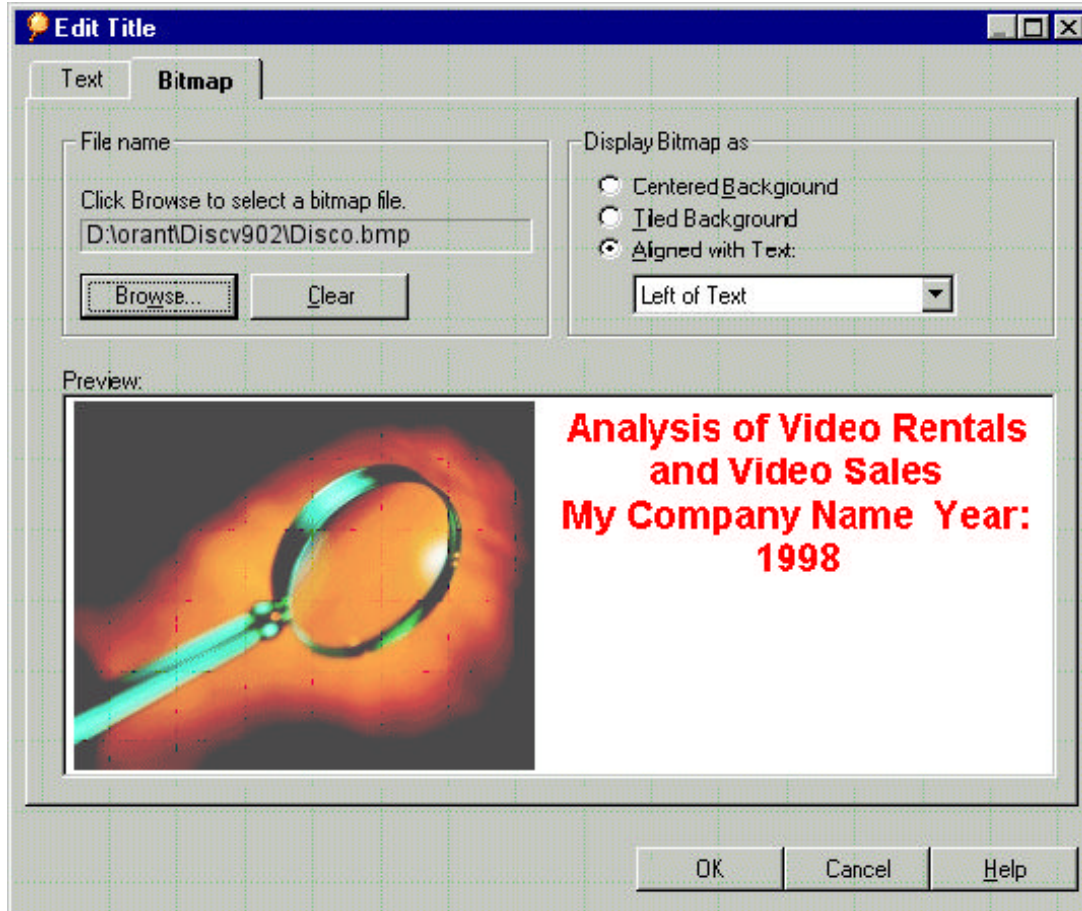
The Title now includes the Year on which the Worksheet is paged.



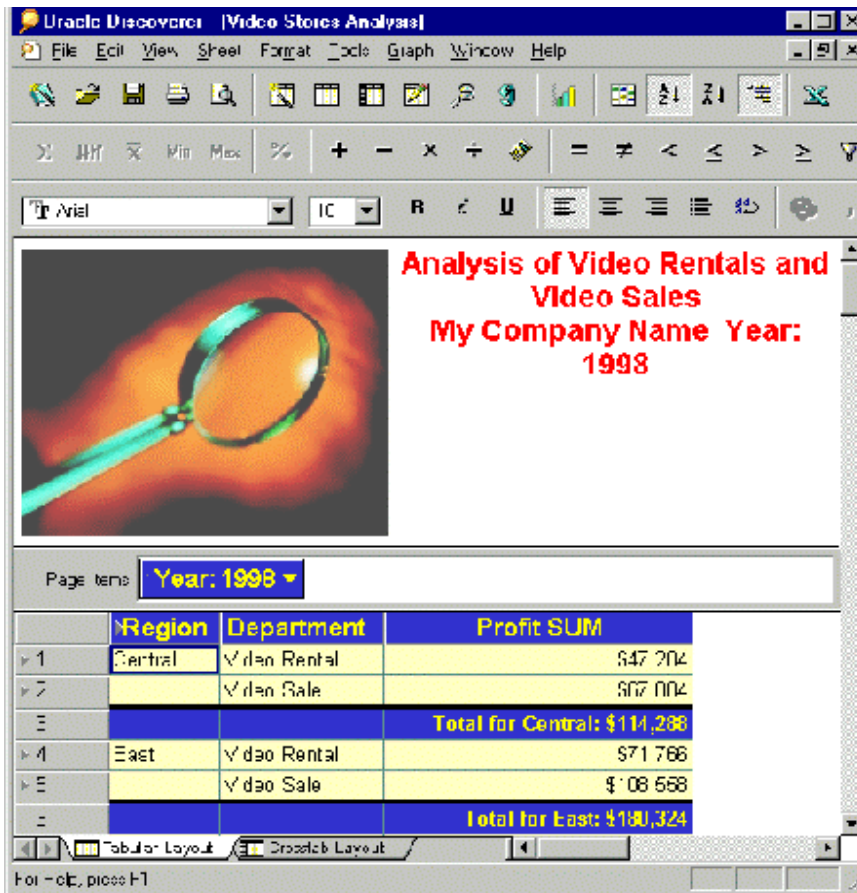
Now that you have edited the title of your report to include your company's name and the Year for which you are reporting. Within this dialog box, you will also add bitmap graphics to your title.

1.3.4 To insert a Bitmap into your Title

1. Choose **Sheet | Edit Title**. The Edit Title dialog box opens.
2. Select the Bitmap tab.
3. Click **Browse** under Bitmap File. An Open dialog box appears.
4. Look in [ORACLE_HOME]\discoverer\ and select **Disco.bmp**, then click **Open**. This file is installed with Discoverer. If you cannot find this file, choose another bitmap (*.bmp) file from your hard drive or network.
5. Under Display Bitmap as, select **Align with Text**.



6. Click **OK** to view the title at the top of your sheet.



Your Worksheet is now personalized for you or your company, with data formatted to make your report clear and easy to analyze. Now, you will begin to analyze your data in order to reveal anomalies in your sales data which will help you make business decisions.

1.3.5 To View a Different Worksheet

You can view the same information from different Worksheet styles within your Workbook. This Workbook has two Worksheets: one Worksheet displays data in a tabular report, which you have already worked with; the other displays data in a crosstabular report.

A crosstab report adds another dimension to your analysis by adding row headings to your report. Discoverer offers you more dimensions through which to view information and refine your analysis—through the use of rows, columns, page Items, and data points. Row headings are positioned along the side axis; column headings are positioned along the top axis; and page Items, at the top of your sheet, are positioned along the page axis. You're now going to view the crosstab Worksheet that has been provided for you in Vidstr5.dis.

From your tabular report, you have seen that the Eastern Region provides the best profit for all your Video Sales. The crosstab Worksheet will enable you to view this information from different angles and to quickly identify areas of the business that are successful, as well as areas that may need help.

1. Click the **Crosstab Layout** tab at the bottom of the screen.

2. If you are asked whether you wish to run this query, click **Yes**.

Analysis of Video Rentals and Video Sales

Page Items:		Department: Video Sale ▼		
		Profit SUM		
		> 1998	> 1999	> 2000
> Central		\$67,084	\$97,921	\$69,493
> East		\$108,558	\$145,462	\$109,637
> West		\$57,096	\$87,172	\$52,092

You are looking at the same annual data for video sales and rentals for all Regions but in a different format.

1.4 Drilling into Data

This crosstab layout shows the profits for each Region for 1998, 1999, and 2000. To see more detailed information about video sales and rentals throughout the Year, you will investigate profits by quarter and month.

Discoverer lets you view different levels of data without starting a new Workbook or Worksheet. This is called **drilling**. You will perform two different types of drilling—an Expand Drill, from Year to Quarter, and a Replacement Drill, from Quarter to Month.

Both the Expand and Replacement drills enable you to drill to different levels of the same data. Later in this tutorial, you will drill to details on related data in Section 1.9, "Drilling to Related Data".

1.4.1 To Drill Down

In our Video Store example, you will investigate profits broken down by quarter.

1. Select the Year Item by clicking on the handle to the left of 1998. A "handle" is the dotted rectangle at the left of the headings along the top axis, or at the top of the headings on the side axis. All three Years, 1998, 1999, and 2000, should be highlighted.

Page Items:		Department: Video Sale ▼		
		Profit SUM		
		> 1998	> 1999	> 2000
> Central	Year	\$67,084	\$97,921	\$69,493
> East		\$108,558	\$145,462	\$109,637
> West		\$57,096	\$87,172	\$52,092

2. Place the cursor on the right-pointing triangle next to the Year **1998**—the cursor changes to a magnifying glass—and click.

A list appears displaying the levels available for viewing. The currently displayed level of information is checked.

Page Items: Department: Video Sale ▾

	Profit SUM		
	2000		
	Calendar Year		
	Calendar Quarter		
	Calendar Month		
Central		,921	\$69,493
East	\$108,558	\$145,462	\$109,637
West	\$57,096	\$87,172	\$52,092

3. Choose **Calendar Quarter**.

Discoverer displays the crosstab report with a breakdown of annual profits by quarter.

Oracle Discoverer - [Video Stores Analysis]

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Analysis of Video Rentals and Video Sales

Page Items: Department: Video Sale ▾

	Profit SUM							
	1998				1999			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	
Central	\$15,531	\$14,734	\$19,053	\$17,766	\$21,670	\$24,768	\$24	
East	\$25,544	\$28,758	\$26,313	\$27,944	\$31,134	\$36,177	\$35	
West	\$13,752	\$16,288	\$14,296	\$12,761	\$20,307	\$21,271	\$22	

Tabular Layout Crosstab Layout

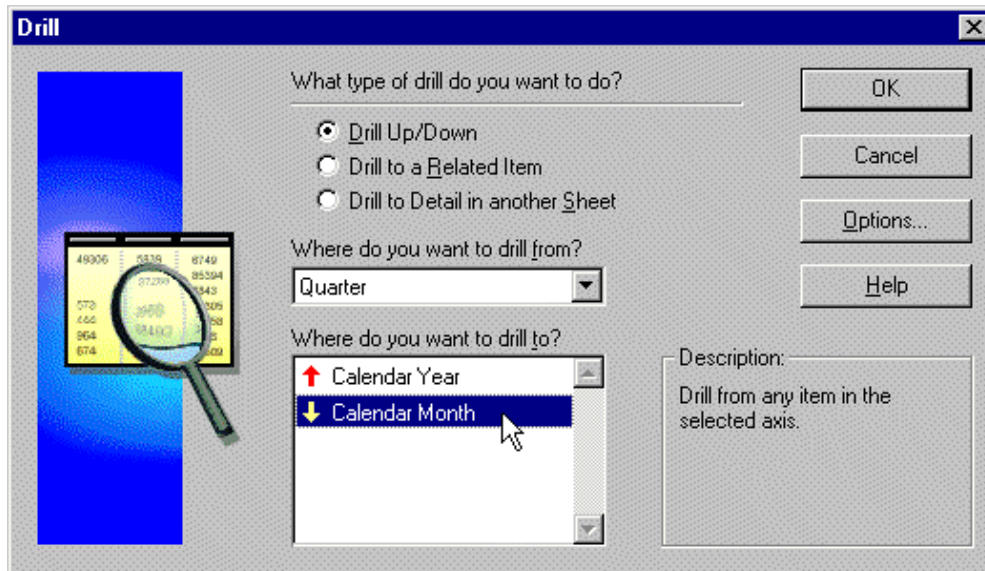
NUM

1.4.2 To Perform a Replacement Drill

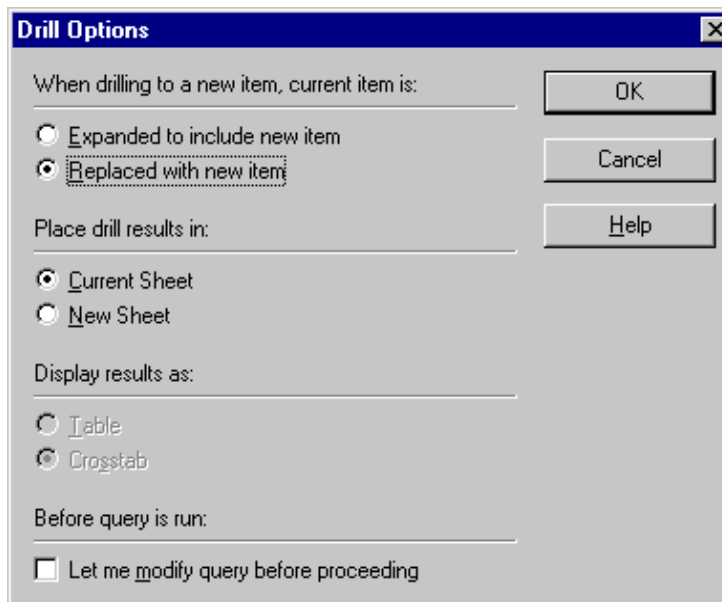
1. Select the Item, **Quarter**, by clicking on the handle. All quarters for all three Years will be highlighted.

	1998		
	Q1	Q2	Q3
Central	Quarter \$15,531	\$14,734	\$19,053
East	\$25,544	\$28,758	\$26,313
West	\$13,752	\$16,288	\$14,296

2. From the main menu, choose **Sheet | Drill**. The Drill dialog box opens.
3. Make sure **Drill Up/Down** is selected.



4. In the **Where do you want to drill to** list, choose **Month**.
5. Click **Options**. The Drill Options dialog box opens.



6. Choose **Replaced with new Item**.
7. Click **OK** in the Drill Options dialog box to close it.
8. Click **OK** in the Drill dialog box to close it and confirm the drill. The data is now displayed by Months within Years.

Oracle Discoverer - [Video Stores Analysis]

File Edit View Sheet Format Tools Graph Window Help

Analysis of Video Rentals and Video Sales

Page Items: Department: Video Sale

	1998	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
Central		\$5,500	\$4,757	\$5,274	\$4,750	\$4,950	\$5,034	\$5,056	
East		\$8,501	\$7,115	\$9,928	\$8,561	\$9,373	\$10,824	\$8,379	
West		\$4,724	\$5,141	\$3,887	\$5,042	\$5,248	\$5,998	\$6,146	

Tabular Layout Crosstab Layout

NUM

Now you can see your profits broken down by month for 1998. Scroll to the right to view the monthly profits for 1999 and 2000.

1.5 Pivoting Data for Comparison and Analysis

It is more useful for you to be able to compare the same quarter for each Year side-by-side. Discoverer makes this very easy. By pivoting data between axes, you can drastically change how you perceive and analyze data in your Worksheet.

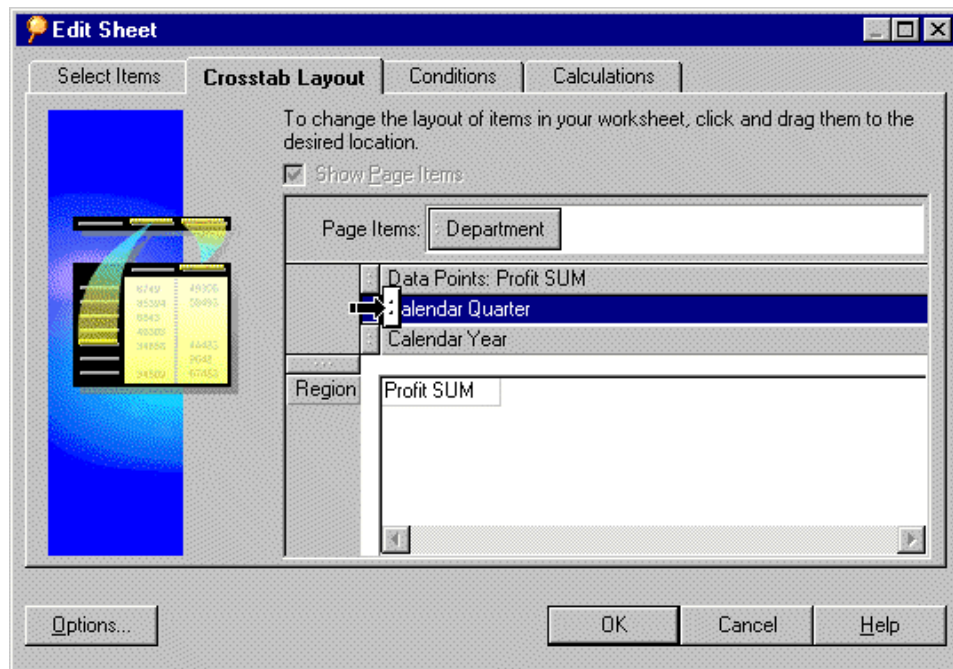
1. First, collapse the month drill data by clicking the handle to the left of the month row, clicking the drill icon, and choosing Calendar Quarter from the pull-down list of drill options. You can now see the quarterly data for each Year.

	1998	Q1	Q2	Mar	Apr	May	Jun
Central		\$5,500	\$4,757	\$5,274	\$4,750	\$4,950	\$5,034
East		\$8,501	\$7,115	\$9,928	\$8,561	\$9,373	\$10,824
West		\$4,724	\$5,141	\$3,887	\$5,042	\$5,248	\$5,998

Calendar Year
Calendar Quarter
✓ Calendar Month

2. Click **Sheet | Edit Sheet** to display the Edit Sheet dialog, and then click the Crosstab Layout tab.

3. Drag the Calendar Quarter Item upwards until it is above the Calendar Year Item, then click **OK**.



You can now easily compare the same quarter for different Years.

Oracle Discoverer - [Video Stores Analysis]

File Edit View Sheet Format Tools Graph Window Help

Analysis of Video Rentals and Video Sales

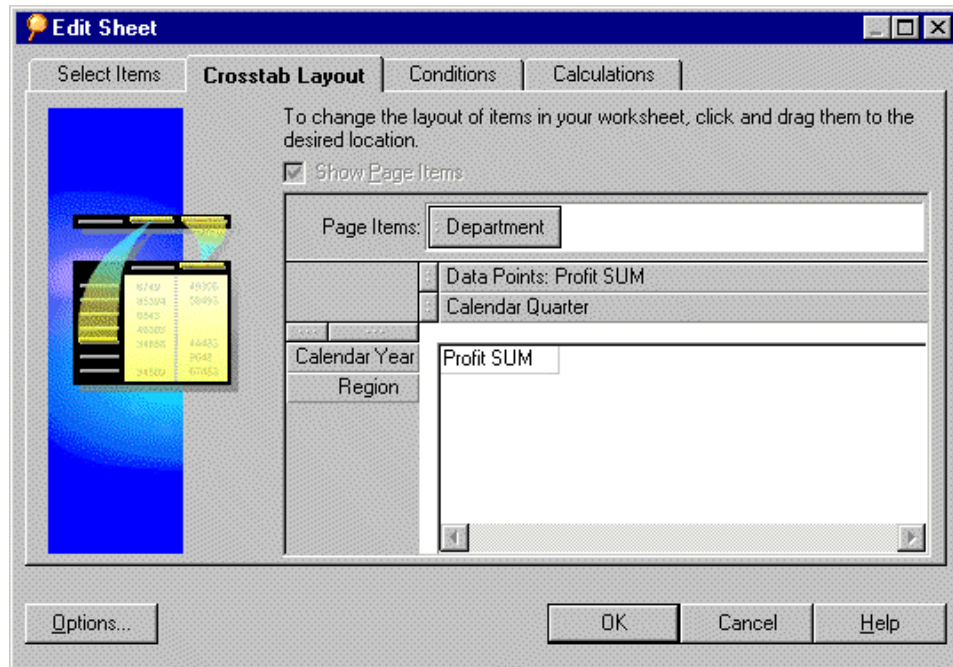
Page Items: Department: Video Sale ▼

	Profit SUM							
	Q1			Q2			Q3	
	1998	1999	2000	1998	1999	2000	1998	1999
Central	\$15,531	\$21,670	\$33,627	\$14,734	\$24,768	\$35,867	\$19,053	
East	\$25,544	\$31,134	\$54,302	\$28,758	\$36,177	\$55,335	\$26,313	
West	\$13,752	\$20,307	\$24,703	\$16,288	\$21,271	\$27,389	\$14,296	

Tabular Layout Crosstab Layout

NUM

4. Click **Sheet | Edit Sheet** to display the Edit Sheet dialog, and then click the Crosstab Layout tab.
5. Drag the Calendar Year Item to the left axis until it is above the Region Item.



6. Click **OK**.

The screenshot shows the Oracle Discoverer Desktop interface with the title bar 'Oracle Discoverer - [Video Stores Analysis]'. The menu bar includes File, Edit, View, Sheet, Format, Tools, Graph, Window, and Help. The toolbar contains various icons for file operations, formatting, and data manipulation. The main area displays a crosstab layout titled 'Analysis of Video Rentals and Video Sales'. The 'Page Items' section shows 'Department: Video Sale' as a dropdown. The data is presented in a crosstab with columns for years (1998, 1999, 2000) and quarters (Q1, Q2, Q3, Q4), and rows for departments (Central, East, West). The values represent profit in dollars.

		Profit SUM			
		Q1	Q2	Q3	Q4
1998		\$54,826	\$59,779	\$59,662	\$58,470
Central		\$15,531	\$14,734	\$19,053	\$17,766
East		\$25,544	\$28,758	\$26,313	\$27,944
West		\$13,752	\$16,288	\$14,296	\$12,761
1999		\$73,110	\$82,216	\$81,993	\$93,236
Central		\$21,670	\$24,768	\$24,181	\$27,302
East		\$31,134	\$36,177	\$35,805	\$42,346
West		\$20,307	\$21,271	\$22,007	\$23,587
2000		\$112,631	\$118,590		

The bottom of the window shows tabs for 'Tabular Layout' and 'Crosstab Layout', with 'Crosstab Layout' currently selected. A 'NUM' button is visible in the bottom right corner.

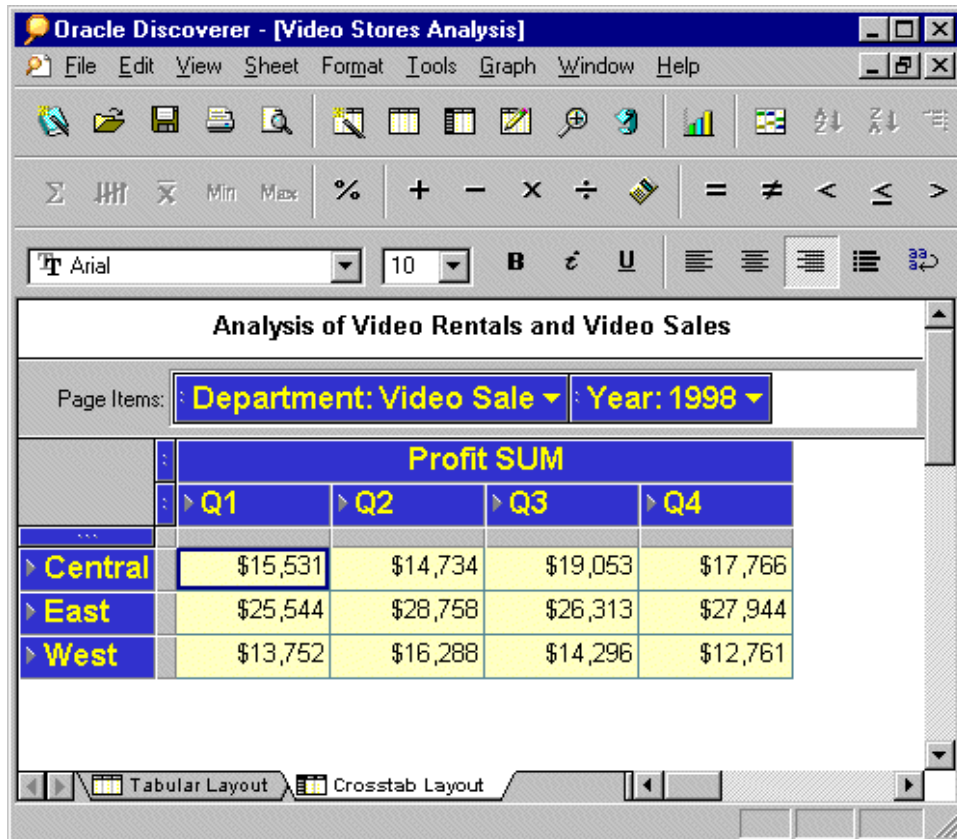
You can now easily compare quarters across different Years.

To simplify the display, you can also view one Year at a time by putting the Calendar Year Item on the Page Items axis.

7. Click **Sheet | Edit Sheet** to display the Edit Sheet dialog, and then click the Crosstab Layout tab.

8. Drag the Calendar Year Item to the Page Items area, to the right of the Department Item.

9. Click **OK**.



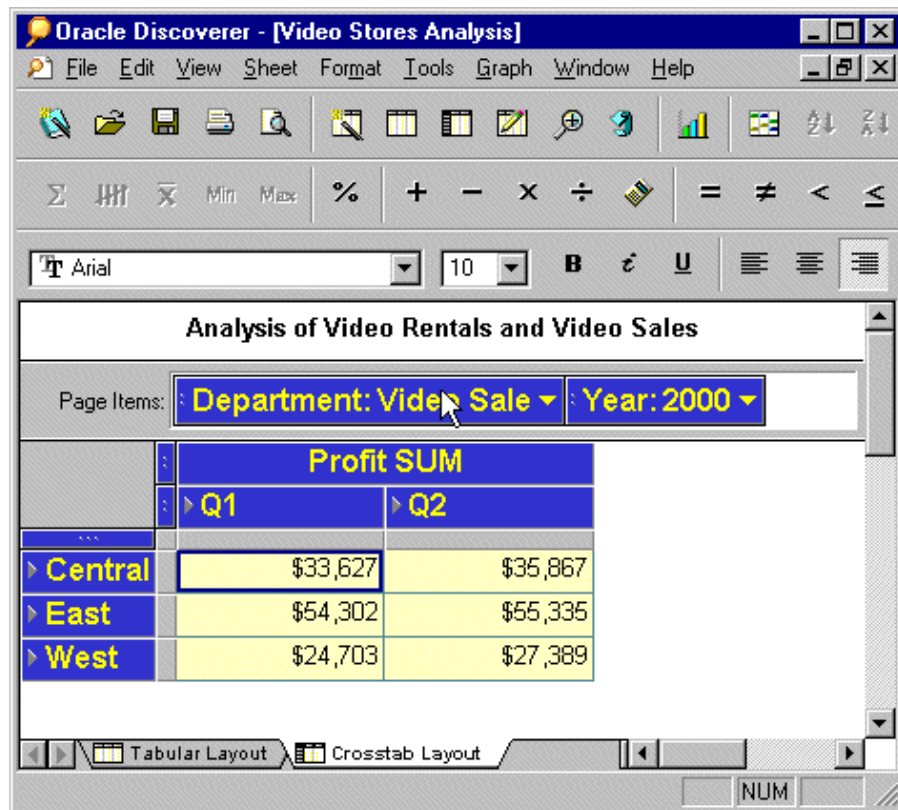
You are beginning to see the power and ease that Discoverer offers you in viewing your data quickly and from many different angles-truly aiding you in analyzing business trends and making decisions.

1.6 Graphing Data Trends

Of course, you will need to report on the results of your business analysis to others. You can use the formatting tools you have learned so far to create clear and functional reports. When you present your findings, you may want to give background information and use a variety of reporting methods.

Discoverer offers you another reporting tool to easily graph your information. In this section you will graph the Video Sale figures in order to compare the profit figures across Regions visually.

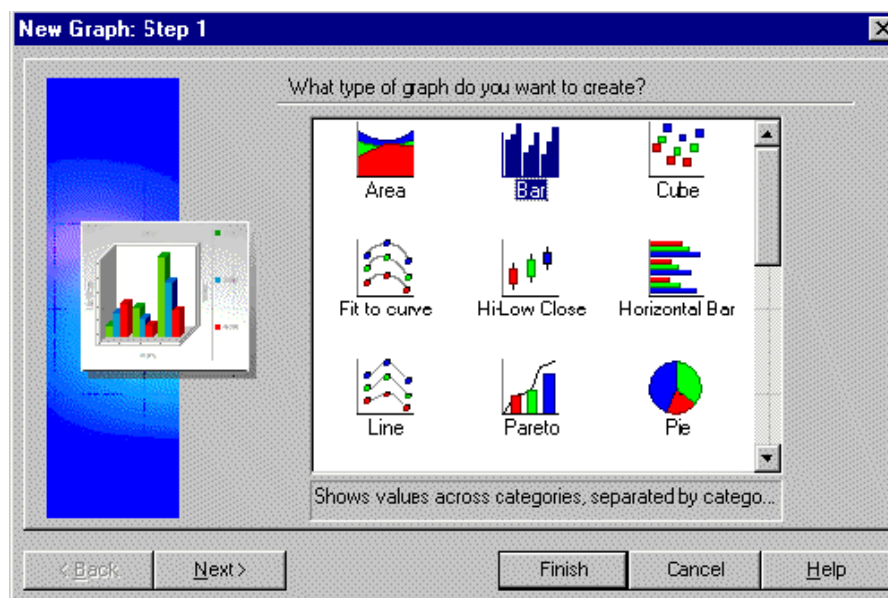
To set up this Worksheet, choose the Crosstab Layout tab from the **Video Stores Analysis** tutorial Workbook. Then choose **Video Sale** from the Department page Item, and **2000** from the Calendar Year page Item, (make sure that Department and Year are displayed in the Page Items area).



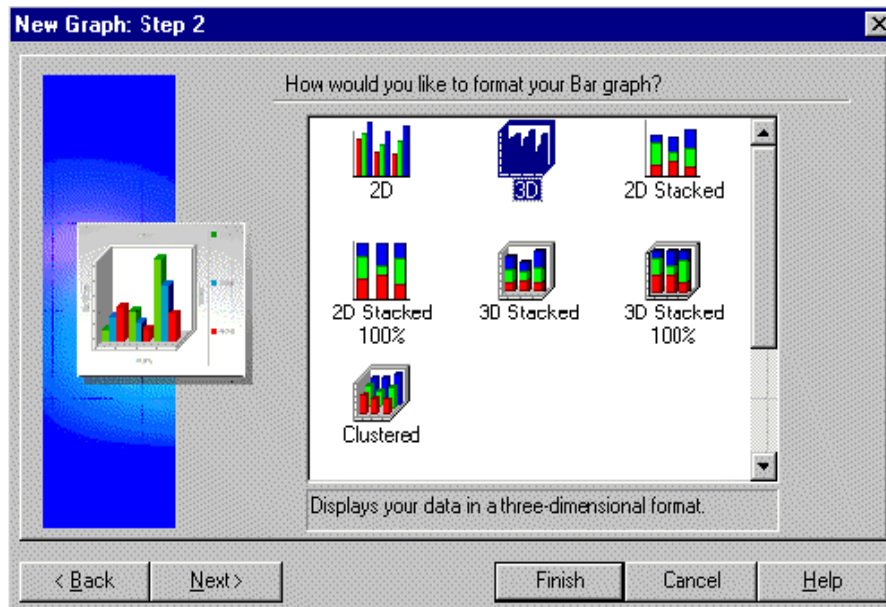
1.6.1 To Create a Graph

1. Click the **Graph** icon  on the toolbar. The New Graph Wizard opens.

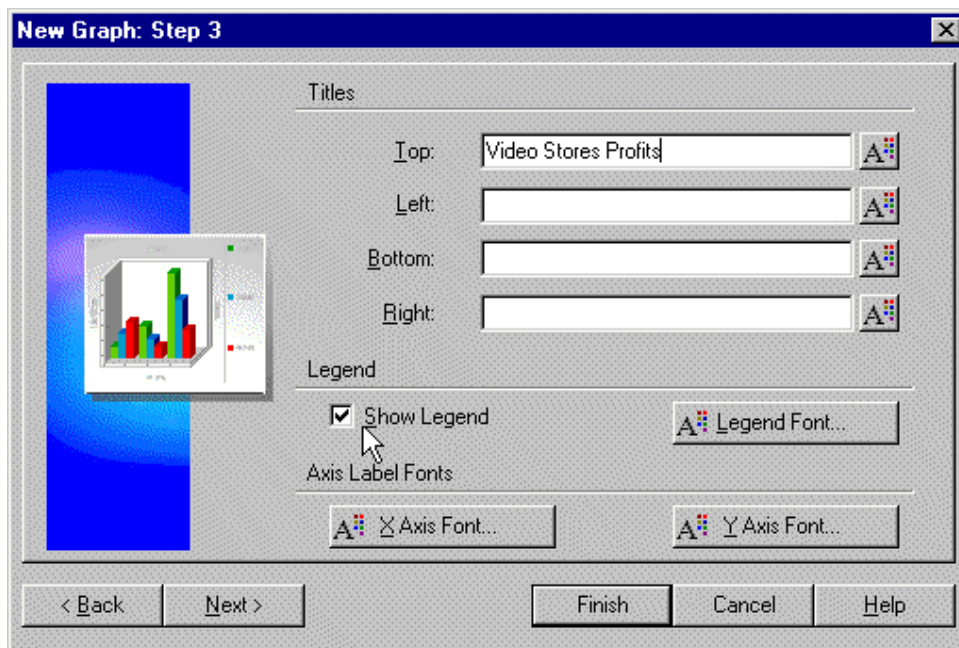
Tip: If you place your cursor on a toolbar icon and leave it there momentarily, the name of the icon pops up.



2. In the gallery of graphs, select **Bar**, and click **Next**. The **New Graph: Step 2** dialog opens.



3. In the gallery of styles, select the **3D** format, and click **Next**. The **New Graph: Step 3** dialog opens.

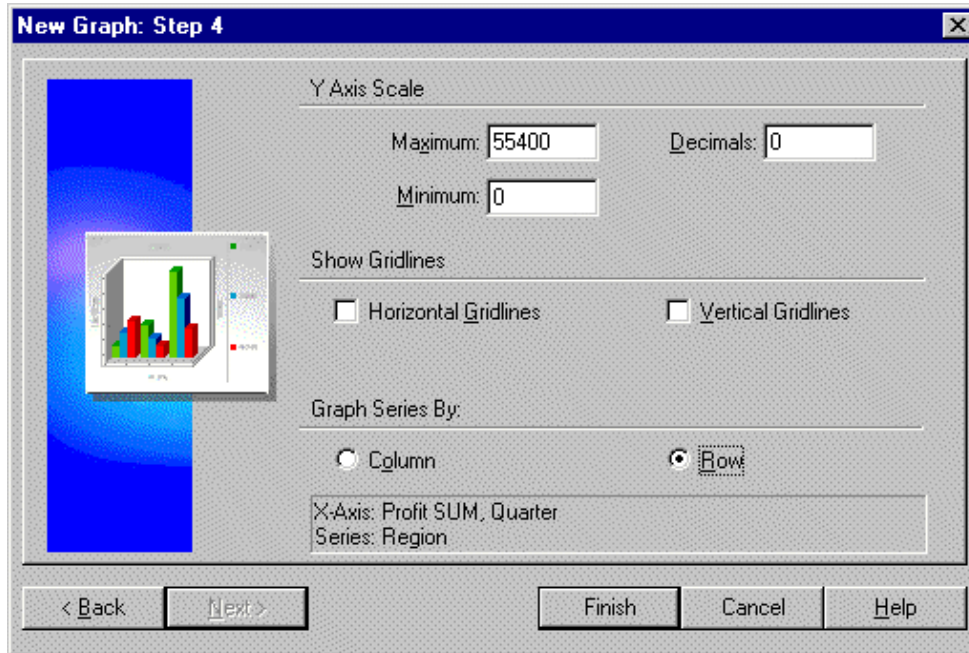


4. You can add titles to any margin of your graph. In the Top field type "Video Stores Profits".

5. Select the **Show Legend** check box.

6. Under "Axis Label Fonts," click **X Axis Font** to format the font for the horizontal axis. You can do the same for the Y axis. Format these fonts at your own discretion.

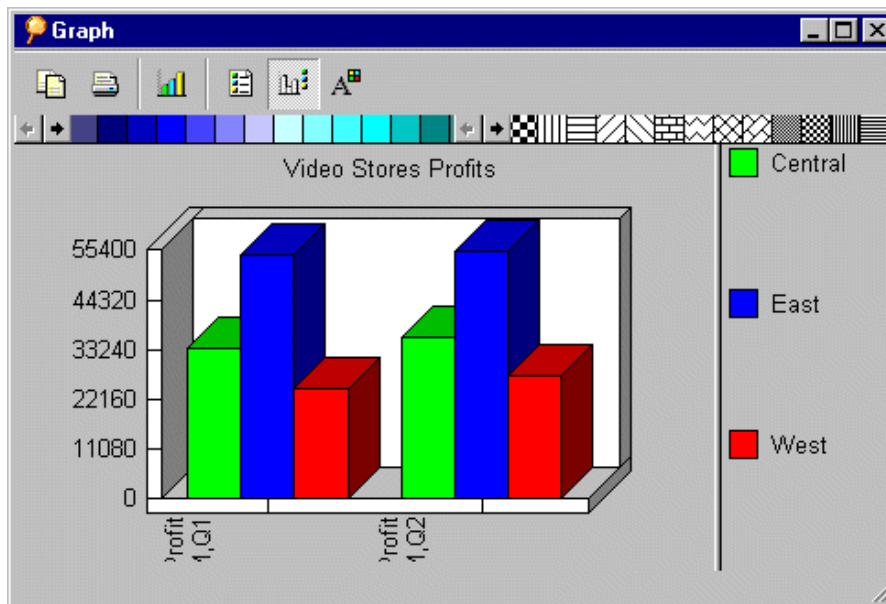
7. Click **Next**. The **New Graph: Step 4** dialog opens.



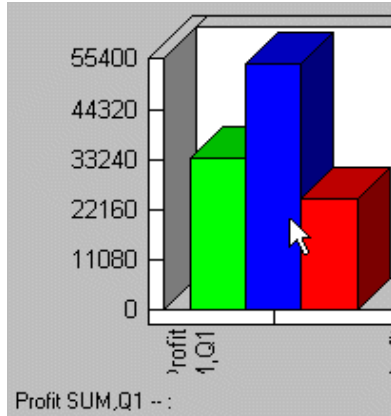
8. Under "Graph Series by" select **Row**.

Notice in the field below that Discoverer reports which data points are on the x-axis and which are the series.

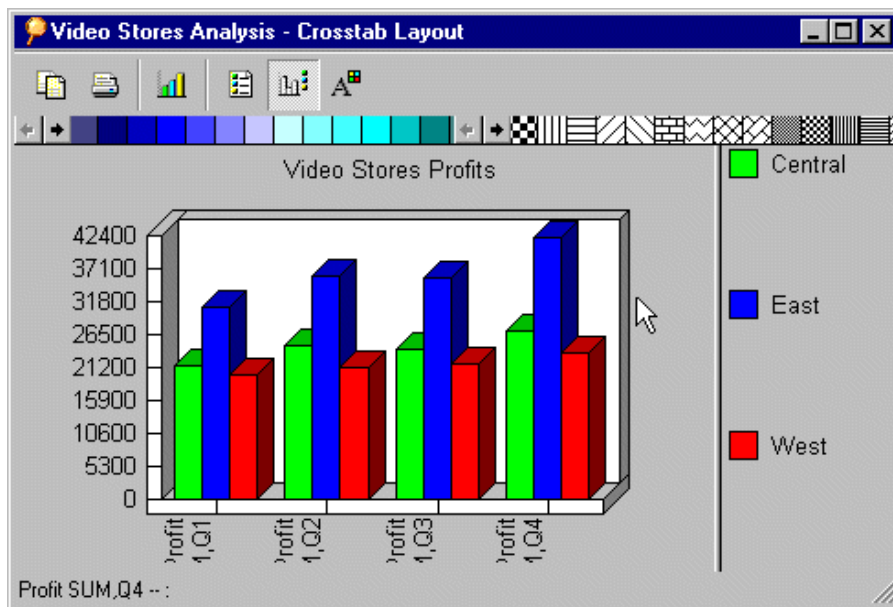
9. Click **Finish**.



This graph shows you the profits for all three Regions for each Quarter in the Year 2000. Notice that as you move your cursor over the bars, the bottom left-hand corner of the panel displays the Item name and Quarter at which you are pointing, (in the example below, **Profit SUM,Q1**).



10. On the Worksheet, change the **Year** page Item from 2000 to 1999. The graph changes to reflect the data. Because we have data for all four Quarters in 1999, the number of series in our graph changes to four.



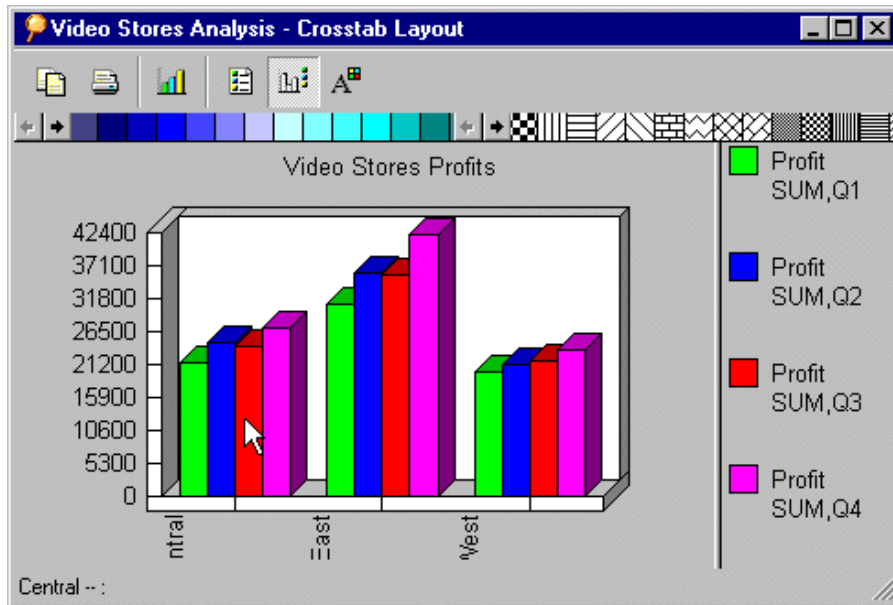
1.6.2 To edit the series of your graph

You are currently viewing Regional profits by Quarter, with the series organized by row. You now want to compare profits across quarters, by grouping the graph columns so that quarter 1 is displayed next to quarter 2 and so on. To do this, we rearrange the **Graph Series By** option to **Column**.

1. Place the cursor in the graph and right click to display a pop-up menu.

2. Choose **Series by Column**.

Tip: You can also choose this option from the **Graph | View** menu Item.



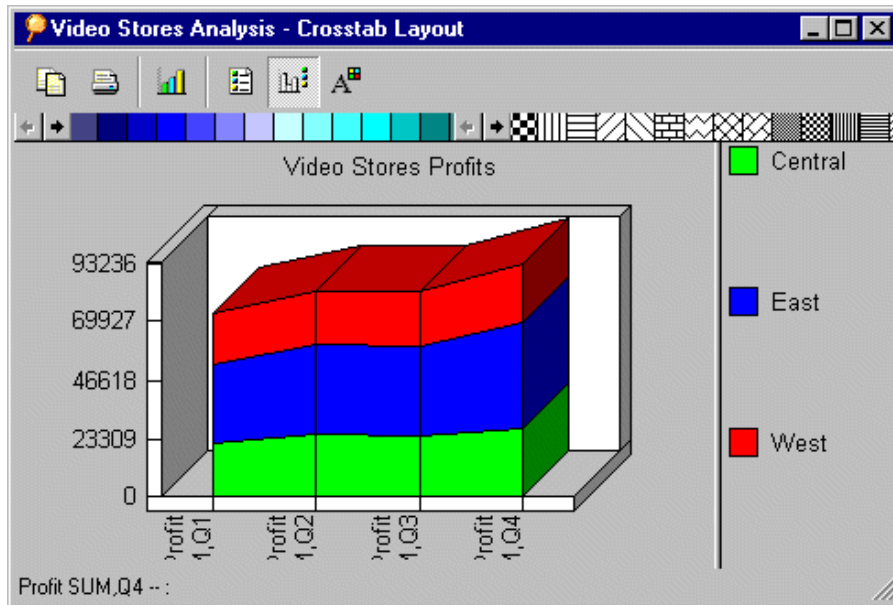
Notice that each Region's monthly profits for the Year are grouped together. This allows you to see an entire Region's profit trends juxtaposed against the others'. You can see that the East Region's profits are considerably higher than those for the other Regions.

3. Repeat Step 1 and choose **Series by Row** to return to the previous display.

1.6.3 To Change the Graph Type and Style

Next, you will explore how to edit the type and style of your graph to expose a different conceptual representation of your data.

1. Select **Graph | Edit Graph**. The Modify Graph dialog box opens.
2. Select the **Type** tab, and choose the Area graph icon from the gallery of graph types (upper left corner).
3. From the **Format** tab, select **3D Stacked**, and click **OK**.



You can now see the profits for each Region stacked on top of each other, with their combined height representing the combined annual profit. Use the cursor to move across each stack to examine the profit values for the individual Region. This kind of graph emphasises the profit for each Region in relation to its contribution to the grand total, more than in comparison to the other Regions.

4. Close the graph by clicking the **Toggle Graph** icon on the toolbar at the top of your Worksheet.

1.7 Printing and Saving your Customized Views

Discoverer lets you preview and print the specific data trends you have isolated in your analysis. Here's how:

- Choose **File | Print Preview** to see how your Worksheet will look when printed.
- Choose **File | Page Setup** to adjust page setups.
- Choose **File | Print** to print the current Worksheet.
- Choose **File | Save** to save changes you have made to the Workbook
- Choose **File | Save As** to save your Workbook to another name. Use this option if you do not want to overwrite the information in the existing Workbook.

Save your Worksheet

You may want to take a break at this time, by saving your Workbook with a unique file name. Throughout this tutorial you may want to take subsequent breaks after finishing a section, then you can return to the tutorial at a later time.

1. Choose **File | Save As**. The Save As dialog box appears.
2. Save your Workbook under your name or another unique name. This will preserve the original Workbook, as well as give you the ability to take a break and come back where you left off.

NOTE: You can save your Workbook in either of two ways:

- 1 - To your local or network drive as a *.DIS file. The Workbook configuration data is stored on

your local or network drive, while the Discoverer data is stored in the Discoverer database.

2 - To the Discoverer database. The Workbook configuration data is stored in the Discoverer database along with the Discoverer data.

1.8 Exposing Exceptions in your Data

The data that you have analyzed in your Video Stores database shows that the profits for the East Region are considerably higher than the other Regions. To see more details about the East, you are going to drill down into the Regional data to see the profits for individual cities.

When analyzing data, you may want to highlight specific pieces of information that fall within or outside a specified range. For example, you may wish to highlight Cities that have profits of more than \$10,000. You can call attention to these records through Discoverer's **Exception Reporting**.

In order to expose anomalous data you will explore monthly profits for each Region and city. You will use Discoverer's Exceptions facility to set up your exception report.

Your Worksheet should be set up in the following manner:

1. In Vidstr5.dis, choose the Crosstab Layout.
 2. **Video Sale** should be selected from the Department Item drop-down list.
 3. Drill down from Year to Quarter.
 4. Place the Calendar Year Item to the right of the Department Item on the Page Axis.
 5. Choose **2000** from the Calendar Year Item drop-down list
 6. Drill down on the **East** Region to **City**.
- (See screen shot below.)

You should be viewing Profit SUM and Quarter on the top axis and Region and City on the left axis.

Oracle Discoverer - [Video Stores Analysis]

File Edit View Sheet Format Tools Graph Window Help

Analysis of Video Rentals and Video Sales

Page Items: Department: Video Sale Year: 2000

	Profit SUM	
	Q1	Q2
Central	\$33,627	\$35,867
East	\$54,302	\$55,335
Atlanta	\$3,590	\$3,715
Boston	\$5,174	\$5,591
Miami	\$1,425	\$1,544
New Orleans	\$3,029	\$3,183
New York	\$21,455	\$20,771
Philadelphia	\$7,236	\$7,875

Tabular Layout Crosstab Layout

NUM

1.8.1 To format exception reports

1. Choose **Format | Exceptions**. The Exceptions dialog box opens.

Exceptions

Define an exception to highlight data in your Worksheet using fonts, colors, and styles. You can define Exceptions for single values or ranges of values.

View Exceptions for: Profit SUM

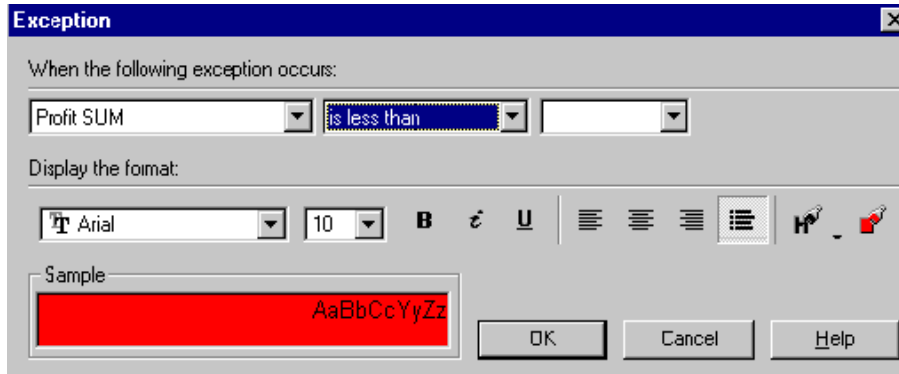
Click New to create a new exception.

Description: No Selection

OK Cancel Help

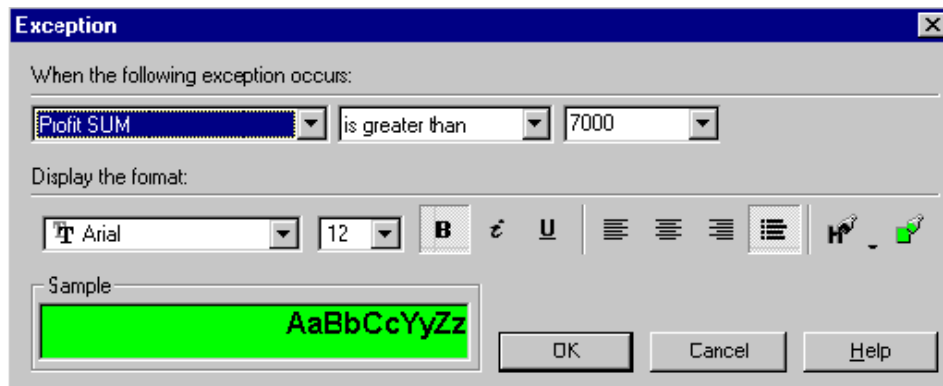
New... Edit... Delete

2. In the "View Exceptions for" drop-down list, choose **Profit SUM** and click **New**. The Exception dialog box opens.



3. To define when the exception occurs, select the following listed parameters from the drop-down lists respectively. You'll need to type the value parameter.

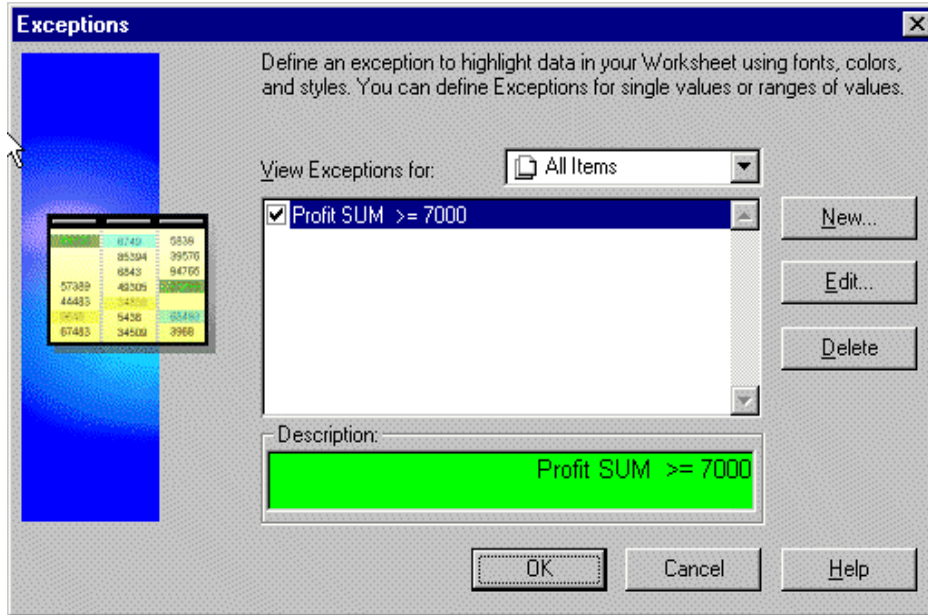
- Profit SUM
- is greater than
- 7000



4. Edit the format display by setting the size to **12**, the style to **Bold**, and the font background to **green**.

The Preview box shows you how your exceptions will look on the Worksheet.

5. Click **OK** to close the Format Exception dialog.



6. Click **OK** to close the Exceptions dialog.

Page Items: Department: Video Sale Year: 2000	
	Profit SUM
	Q1 Q2
Central	\$33,627 \$35,867
East	\$54,302 \$55,335
Atlanta	\$3,590 \$3,715
Boston	\$5,174 \$5,591
Miami	\$1,425 \$1,544
New Orleans	\$3,029 \$3,183
New York	\$21,455 \$20,771
Philadelphia	\$7,236 \$7,875
Pittsburgh	\$5,892 \$6,477
Washington	\$6,501 \$6,179

You can now see at a glance which cities have profits greater than \$7,000, highlighted in green. As you can see, New York and Philadelphia show outstanding profits, and that New York is the leading store in the Eastern area. It is obvious that this store has contributed to the overall success of the East Region.

Note that the summary profit values for the three Regions all exceed the Exception amount, and are therefore also highlighted.

- Now is a good time to save your Workbook if you would like to take a break. The next section of the tutorial will continue from this point.

1.9 Drilling to Related Data

Using Discoverer's exception reporting capability, you have highlighted an unusually high increase in profit for New York for the quarters one and two, (Q1 and Q2) in the Calendar Year 2000. You can investigate further to find out the cause of this dramatic increase through Discoverer's ability to drill to detailed or related information.

Your Worksheet should be set up in the following manner:

1. In **Video Stores Analysis**, choose the Crosstab Layout.
2. **Video Sale** should be selected from the Department Item drop-down list.
3. **2000** should be selected from the Calendar Year Item drop-down list
4. Drill down on the **East** Region to **City**.

Page Items:		: Department: Video Sale ▾		: Year: 2000 ▾	
		:	Profit SUM		
		:	> Q1	> Q2	

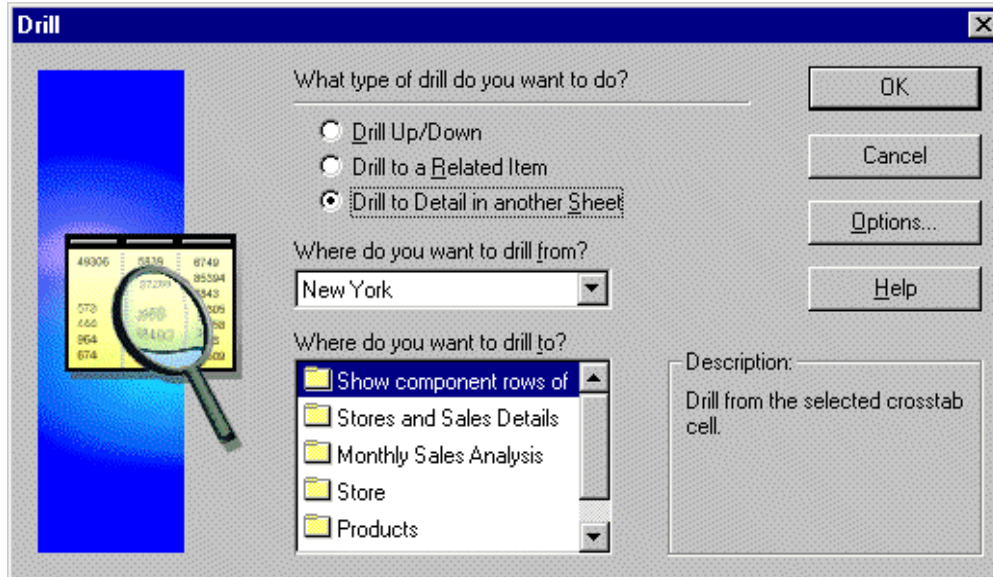
> Central			\$33,627	\$35,867	
> East			\$54,302	\$55,335	
> Atlanta			\$3,590	\$3,715	
> Boston			\$5,174	\$5,591	
> Miami			\$1,425	\$1,544	
> New Orleans			\$3,029	\$3,183	
> New York			\$21,455	\$20,771	
> Philadelphia			\$7,236	\$7,875	
> Pittsburgh			\$5,892	\$6,477	
> Washington			\$6,501	\$6,179	

To drill to related details

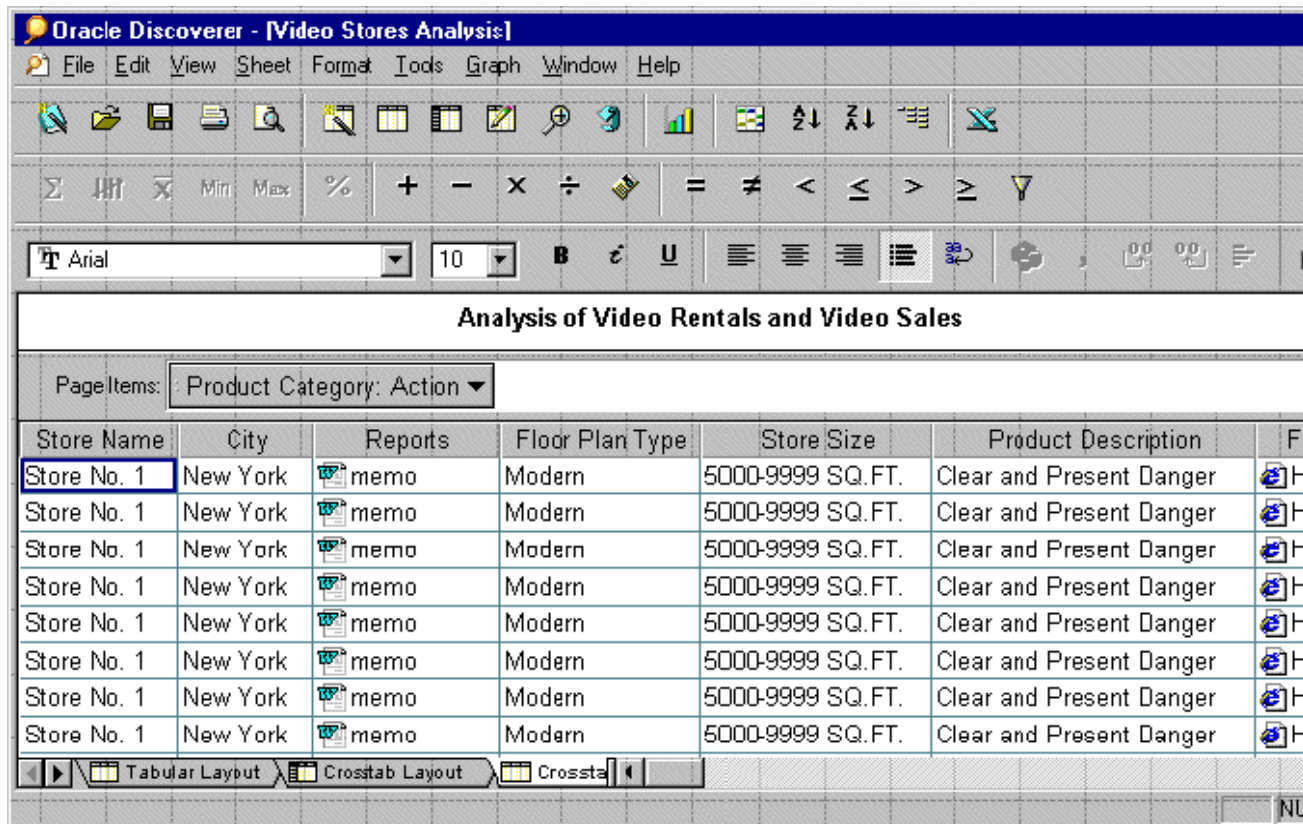
1. Double-click the cell that represents New York's profits for quarter one (Q1), 2000 (\$21,455).

East		\$54,302		\$55,335	
Atlanta		\$3,590		\$3,715	
Boston		\$5,174		\$5,591	
Miami		\$1,425		\$1,544	
New Orleans		\$3,029		\$3,183	
New York		+	\$21,455		\$20,771
Philadelphia		\$7,236		\$7,875	

The Drill dialog box appears.



2. Select **Drill to Detail in another sheet**.
3. Double click the **Store and Sales Details** folder.



A new Worksheet is created with the details of video sales for New York in quarter one in 2000. This Worksheet contains reports and memos explaining why New York's profits were so high.

4. To drill out to one of the Microsoft Word **memo** reports, double-click on one of the Microsoft

Word icons in the Reports column.

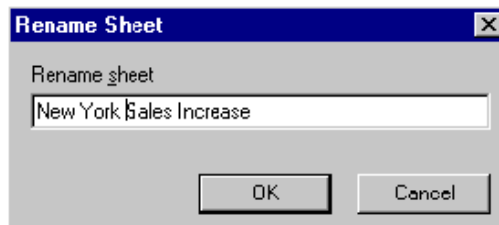
Page Items: Product Category: Action ▼			
Store Name	City	Reports	Floor Plan Type
Store No. 1	New York	memo +	Modern
Store No. 1	New York	memo	Modern
Store No. 1	New York	memo	Modern

The selected report will be displayed in Microsoft Word or Viewer application installed on your computer. You can also double click on the Internet Explorer icons in the Full Description column to drill out to a Web site.

1.10 Renaming a Worksheet

It is easy to rename your new Worksheet by simply double-clicking its tab.

1. Double-click the Worksheet's tab at the bottom of the Workbook page. The **Rename Sheet** dialog box opens.



2. Type **New York Sales Increase**.

3. Click **OK**.

This is a good time to save your Workbook. The new Worksheet will be saved with it.

1.12 Exporting your Data to other Formats

You can export your Worksheets to numerous formats including HTML (*.HTM), Text (*.TXT), Lotus 1-2-3, Microsoft Excel. This feature is helpful in distributing your reports electronically to others who don't have Discoverer.

1.12.1 To Export your Data as a .HTML File

In this section we export the Tabular Layout Worksheet of the Vidstr5.dis Workbook to a document that can be viewed from your web browser.

1. Click on the Tabular Layout tab in the Vidstr5.dis Workbook.

Oracle Discoverer - [Video Stores Analysis]

File Edit View Sheet Format Tools Graph Window Help

Analysis of Video Rentals and Video Sales

Page Items: Year: 1998

	Region	Department	Profit SUM
1	Central	Video Rental	\$47,204
2		Video Sale	\$67,084
3			Total for Central: \$114,288
4	East	Video Rental	\$71,766
5		Video Sale	\$108,558
6			Total for East: \$180,324
7	West	Video Rental	\$39,395
8		Video Sale	\$57,096
9			Total for West: \$96,491
10			Total for All Values: \$391,104

Tabular Layout Crosstab Layout

NUM

2. Choose **File | Export**. The Export dialog box appears.

Export

What do you want to export?

☒ Current Sheet
☐ All Sheets
☐ Selection
☐ Graph

What export format do you want?

Hyper Text Markup Language (*.htm)

Where do you want to save the file?

D:\orant\Discv902\Video Stores Analysis.htm

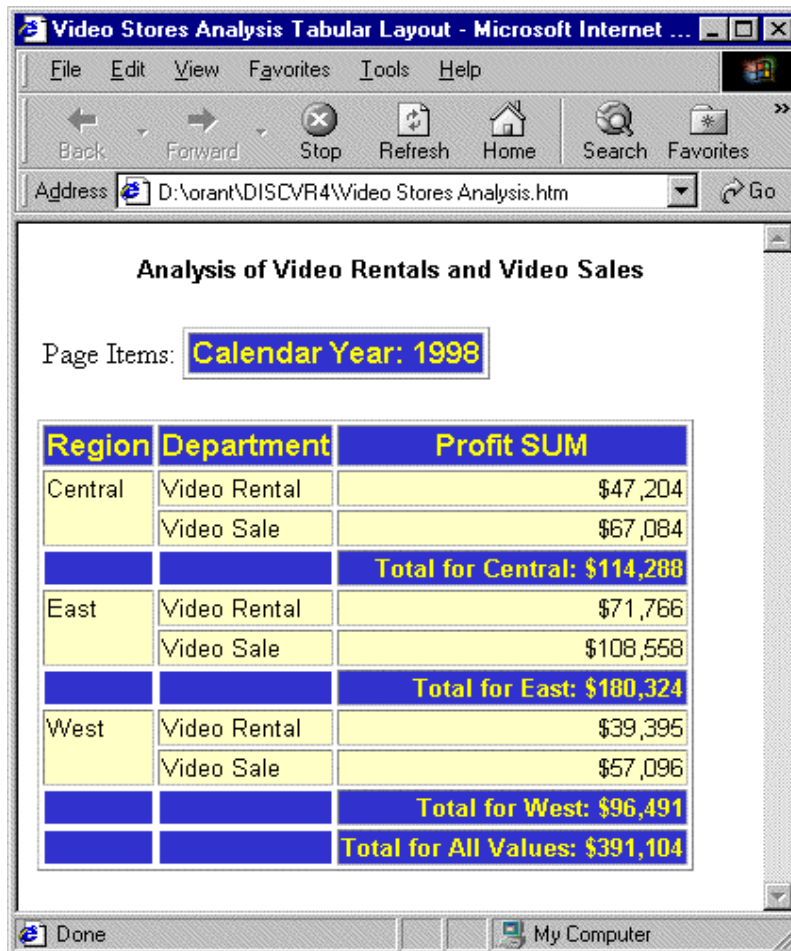
Browse...

Finish Cancel Help

3. Choose to export **Current Sheet**, and choose **Hyper-Text Markup Language (*.htm)** as the export format.
4. Select a directory in which to save the exported file. The files will be named automatically, but you can change the file name as required.
5. Click **Finish**.

Discoverer saves the current Worksheet to an .htm file in the directory you specified.

When you open the exported file in your web browser, you will see your Discoverer data replicated in HTML format.

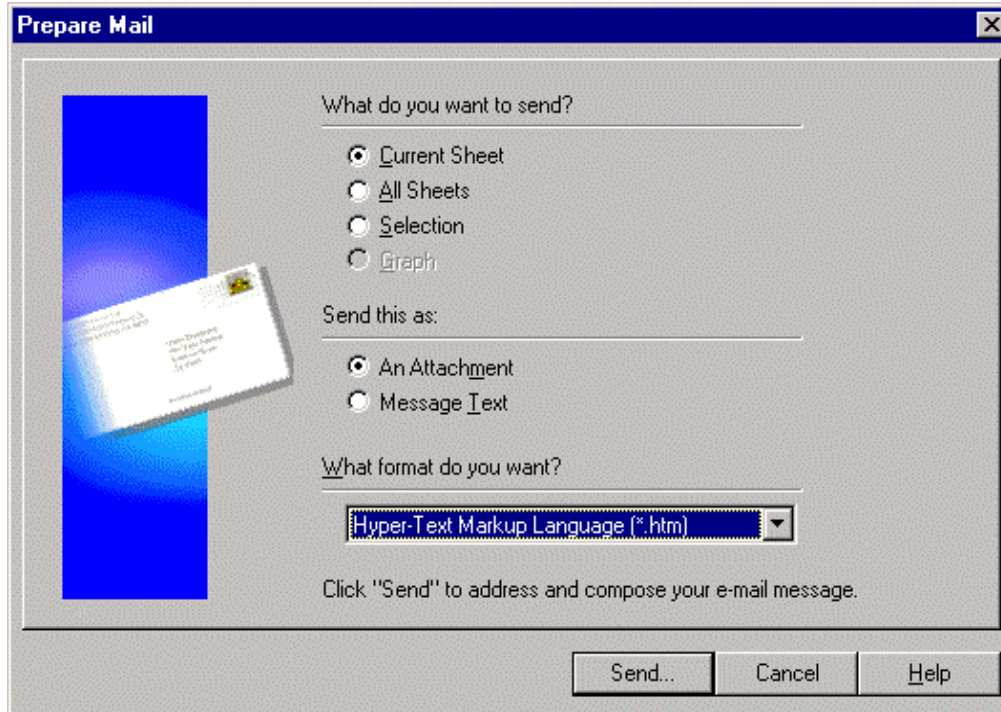


1.12.2 To E-Mail your report as an Attachment

If your e-mail application supports sending messages and attachments from other applications, you can e-mail Discoverer Worksheets.

Note: If the **File | Send** menu Item is not available, then your e-mail application does not support this function, and you should skip to the next section, Section 1.13, "Creating a Workbook".

1. Choose **File | Send**. The Prepare Mail dialog box appears.



2. Click on **Current Sheet** and **An Attachment**, and select **Hyper-Text Markup Language (*.htm)** from the pull-down list.

3. Click **Send**.

Note: From this point on, you will not be working from Discoverer, but from the Microsoft interface and your e-mail package, so there may be some variation.

The Choose Profile dialog box may appear if you have more than one e-mail profile installed on your machine. Select your e-mail package from the pull-down list in the Choose Profile dialog box.

Discoverer will start your e-mail program, if it is not already running, and open a New Message.

Notice the attachment is already a part of your message. Its name is the Workbook name plus Worksheet name. The subject has also been filled in for you with the name of the document you are attaching; you may change this if you wish.

4. Address and send your e-mail to yourself or a co-worker.
Later on, you can go to your e-mail and check your message.

Note: For more information on sending Discoverer reports in other file formats, refer to the Discoverer Desktop User's Guide.

Congratulations! you have mastered many of the features of Discoverer which assist you in your business analysis and decision making. You know how to edit and format Worksheets, revealing meaningful information in a report that is easy to view and distribute. You may want to save your Workbook at this time, before going on to creating your own Workbook from scratch.

1.13 Creating a Workbook

So far you have been working with a Workbook (**Video Stores Analysis**) that someone else created. However, you may need to create your own workbook to analyze different information. You do this using the Workbook wizard.

In this section, you will complete the following steps involved in creating a new Workbook:

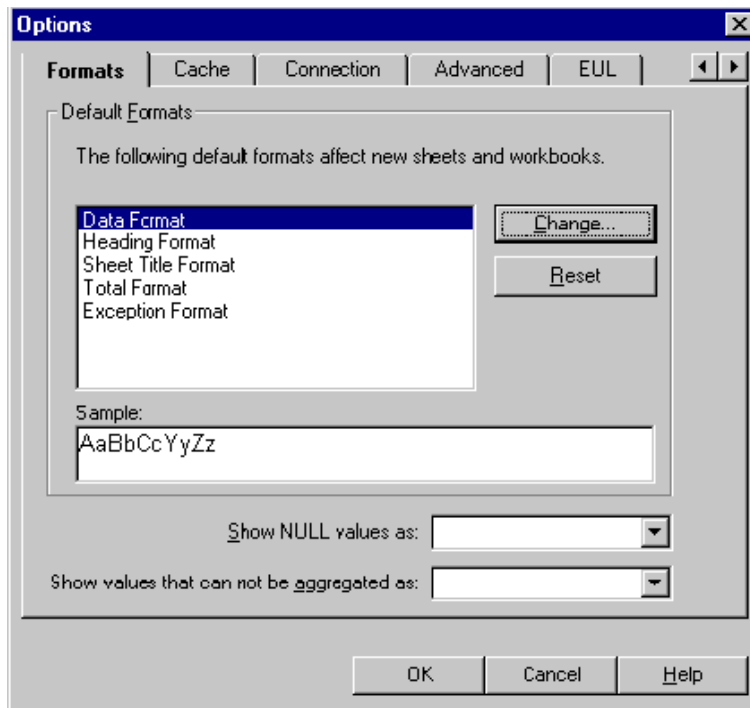
- Set up default formats for all new Worksheets
- Choose a layout for displaying your information
- Choose an area from which to gather your information
- Choose the pieces of information you want to view
- Choose the layout of your information

1.13.1 To change default formats

Before you begin to create your own Worksheet, you are going to set up default formats according to your own preferences. You may keep your previous Workbook open. You will not see changes to this previously created Workbook, as these default formats will only effect Workbooks created after the formats have been defined.

1. Choose **Tools | Options**.

The Options dialog box opens.



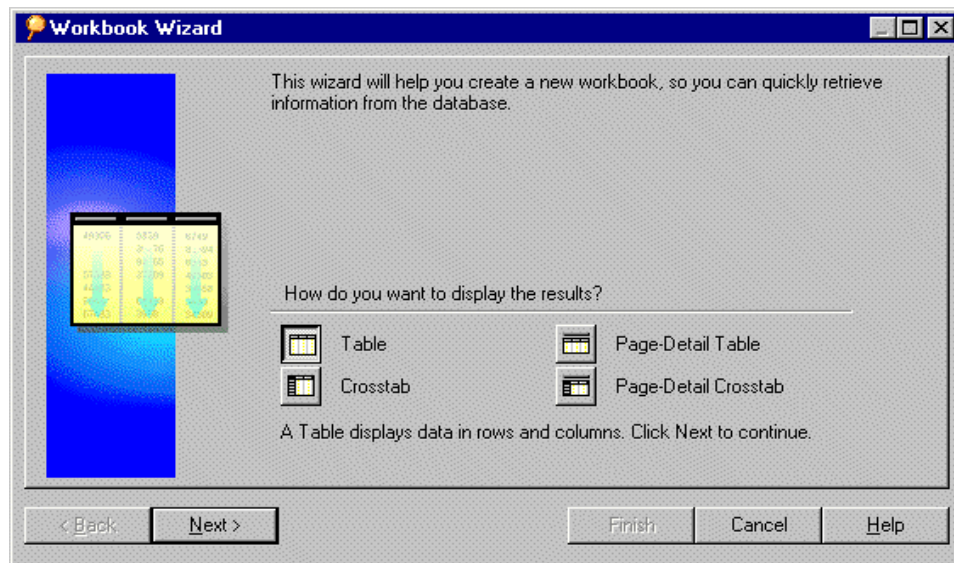
2. Select the **Formats** tab.
3. In the Default Formats list, select **Data Format**.
4. Click **Change**. The Format Data dialog box opens.

5. Define the appearance of your data to your own preferences by changing the Font, Size, Style, and Color.
Use the Sample box to preview your changes.
6. Select the Background Color tab and choose a background color.
7. Click **OK** to close the Format Data dialog box.
8. Click **OK** to close the Options dialog box and apply your changes.
These settings will be seen when you create your new Workbook. To change the default format of a heading, Worksheet title, total, or exception, follow the above steps, selecting the appropriate format from the Default Formats list (Step 4).

1.13.2 To create a Worksheet using the Workbook Wizard

In our video store example, you will investigate the total profit across Regions and Departments to find the Regions that are performing exceptionally well.

1. Choose **File | New** to start the **Workbook Wizard**.

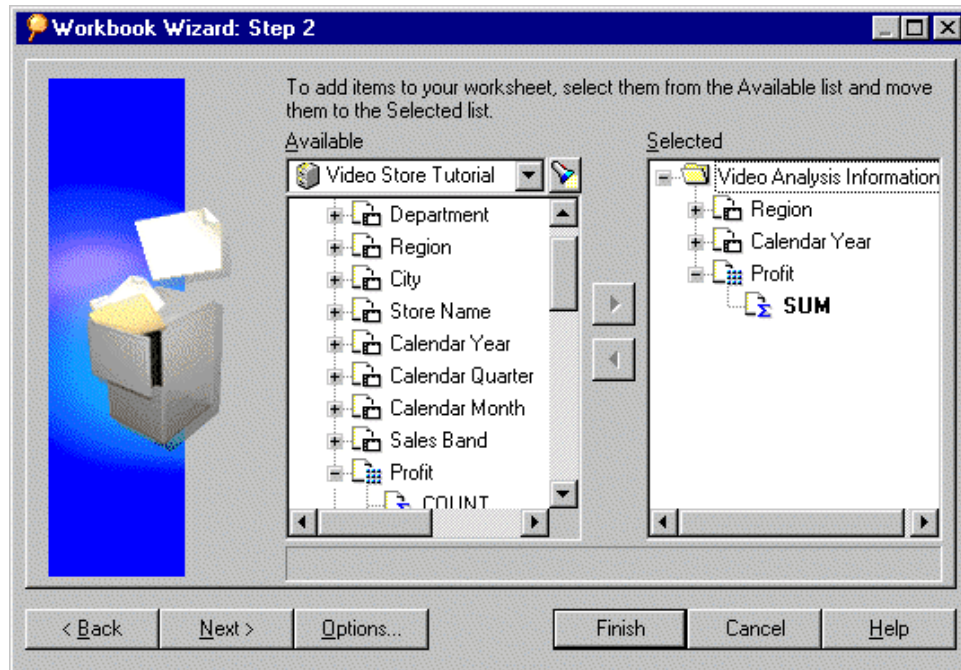


2. Click **Table** from the **How do you want to display the results** options.
3. Select **Next** to display **Workbook Wizard: Step 2**.

Choose a Business Area and data for your report.

In **Workbook Wizard: Step 2**, you will choose the Business Area and data for your report. The **Available** list tells you all of the Business Areas your Discoverer manager has set up for you to access. In the panel below, you can see a list of the folders available to you as part of the Video Store Tutorial Business Area.

1. In **Workbook Wizard: Step 2**, Choose Video Store Tutorial from the **Available** list.



2. Expand the **Video Analysis Information** folder by clicking the + next to the folder.

3. Select the following Items by clicking and dragging them to the **Selected** area:

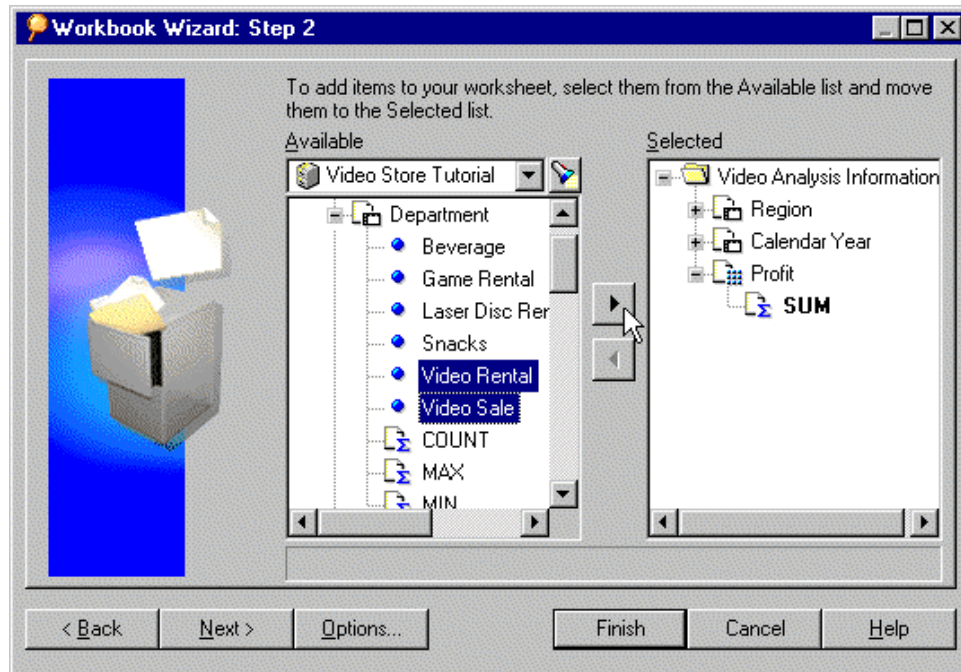
- Profit SUM
- Region
- Calendar Year

Tip: Use the scroll bar to view all of the Items, or resize this dialog box by clicking and dragging the outer edges to see more information.

4. Expand the Department Item, then select and drag the following Items to the Selected list:

- **Video Rental**
- **Video Sale**

Tip: Hold down the Ctrl key to select multiple Items.



5. Click **Next**. The **Workbook Wizard: Step 3** opens.

Modify the layout of your information

In **Workbook Wizard: Step 3**, Discoverer enables you to modify the layout of your information before it gathers that information.

6. Move Profit SUM to the right of Region.

7. Move Calendar Year to the left of Region.

8. Click **Finish**.

9. Click **Format|Column|Autosize** to set the column width, or adjust them manually by dragging the column borders.

You have now created a Workbook with one Worksheet that shows you the profit by Region for Video Sales and Rentals for 1998, 1999, and 2000.

Page Items: Department: Video Rental ▼		
Year	Region	Profit SUM
1998	Central	\$47,204.15
1998	East	\$71,766.31
1998	West	\$39,395.22
1999	Central	\$43,812.58
1999	West	\$37,777.44
1999	East	\$67,248.68
2000	Central	\$25,157.29
2000	West	\$23,521.02
2000	East	\$40,401.56

1.14 Performing Calculations

Discoverer makes it extremely easy for you to perform a variety of calculations yourself. This section will teach you to set up a variety of calculations, such as simple totals, percentages, and averages, for your Video Store business. It will also introduce a couple of features that will help you view and arrange your data to aid your analysis.

1.14.1 Sorting your Data

You can set additional conditions and sorts, as well as create calculated Items, by using the remaining pages of the Workbook Wizard. In your Video Store analysis Worksheet, you are going to sort by Region and add formatting to view your information more clearly.

1. Right-click on the **Calendar Year** and choose **Group Sort**.

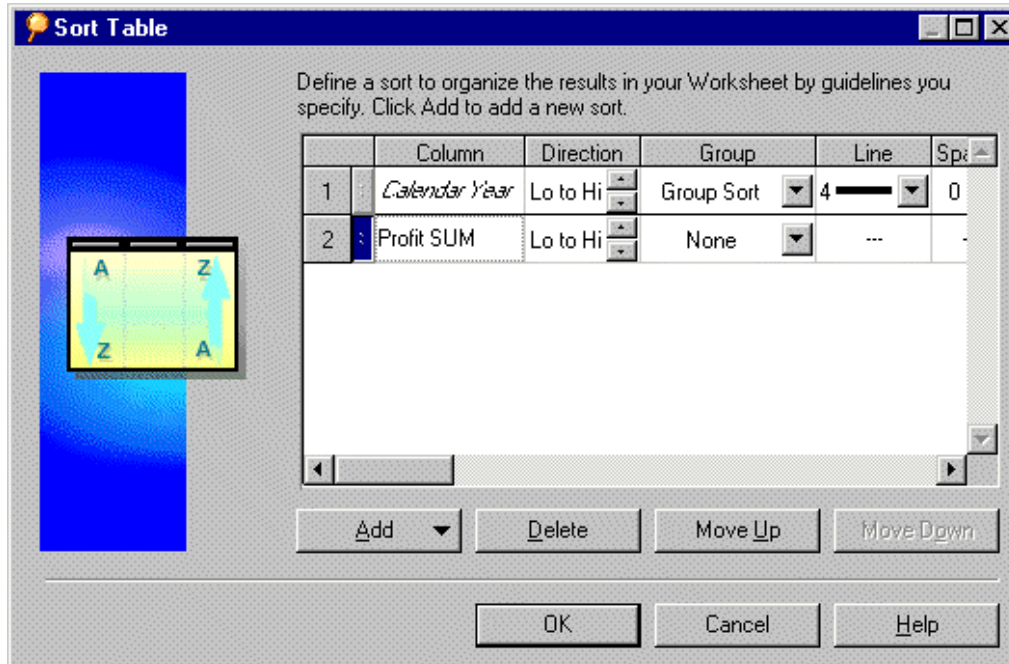
Page Items: Department: Video Rental ▼		
Year	Region	Profit SUM
199	Item Properties...	\$47,204.15
199	Cut	\$71,766.31
199	Copy	\$39,395.22
199	Delete	\$43,812.58
199	Format Data...	\$37,777.44
199	Format Heading...	\$67,248.68
200	Format Exception...	\$25,157.29
200	Column Width...	\$23,521.02
200	Column Auto Size	\$40,401.56
	Group Sort	
	Sort Low to High	
	Sort High to Low	
	Show Values...	

Now you see your information sorted by Year. Next, make your information easier to view by adding a secondary sort by Profit SUM.

Page Items: Department: Video Rental ▼

Year	Region	Profit SUM
1998	Central	\$47,204.15
	East	\$71,766.31
	West	\$39,395.22
1999	Central	\$43,812.58
	West	\$37,777.44
	East	\$67,248.68
2000	Central	\$25,157.29
	West	\$23,521.02
	East	\$40,401.56

2. Select **Tools | Sort**. The Sort Table dialog box opens.



3. Use the drop down list to change the line width to 4 for Calendar Year. This will make it easier to distinguish each Region's data.

4. Click **Add** and select **Profit SUM**. This adds Profit SUM as a secondary sort.

5. Click **OK**.

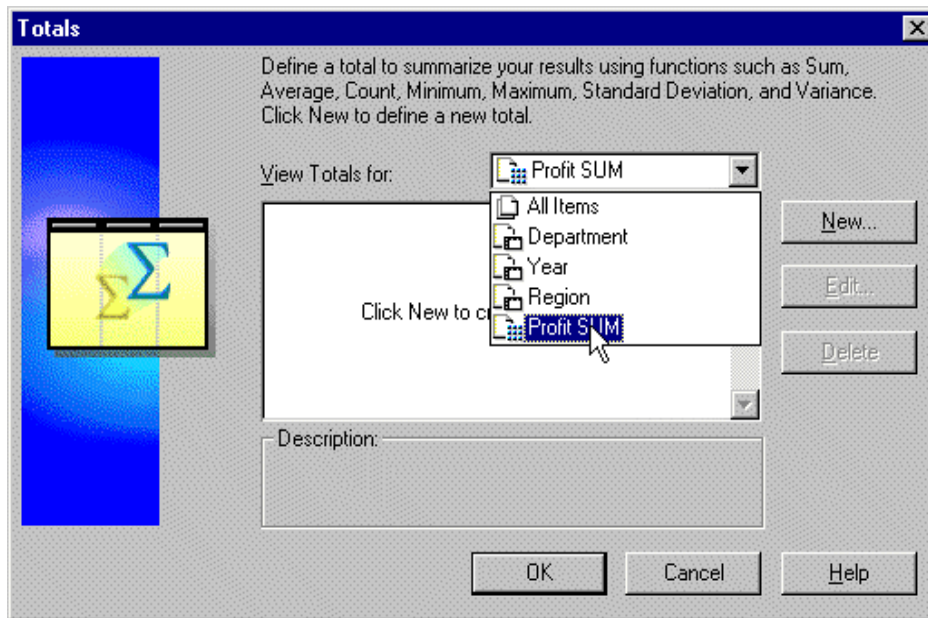
Page Items: Department: Video Rental ▼		
Year	Region	Profit SUM
1998	West	\$39,395.22
	Central	\$47,204.15
	East	\$71,766.31
1999	West	\$37,777.44
	Central	\$43,812.58
	East	\$67,248.68
2000	West	\$23,521.02
	Central	\$25,157.29
	East	\$40,401.56

Your Worksheet is now sorted by Profit SUM within Calendar Year. Increasing the line width makes it easier to differentiate the data for each Year.

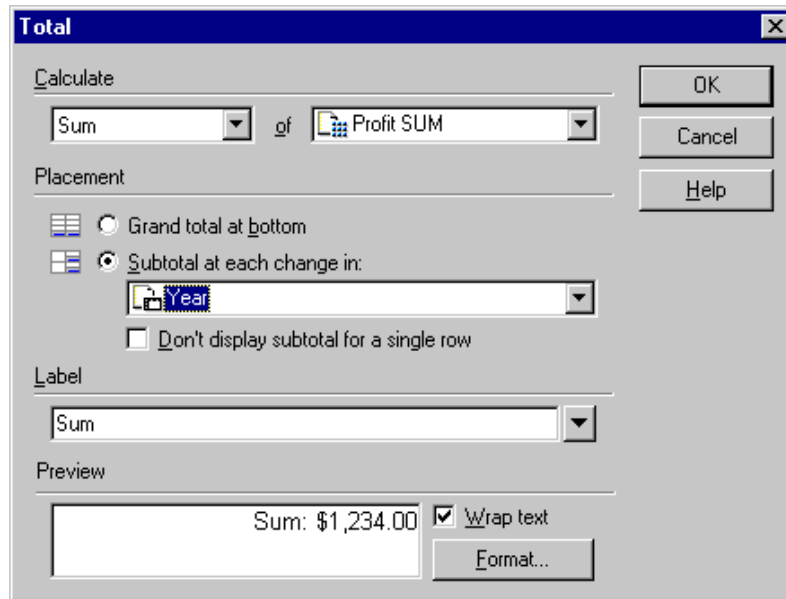
1.14.2 To Calculate Total Profit

Now that the data for each Year is distinguishable and easy to view, you will subtotal the profit for each Year. This will give you annual profit figures for the three Regions combined.

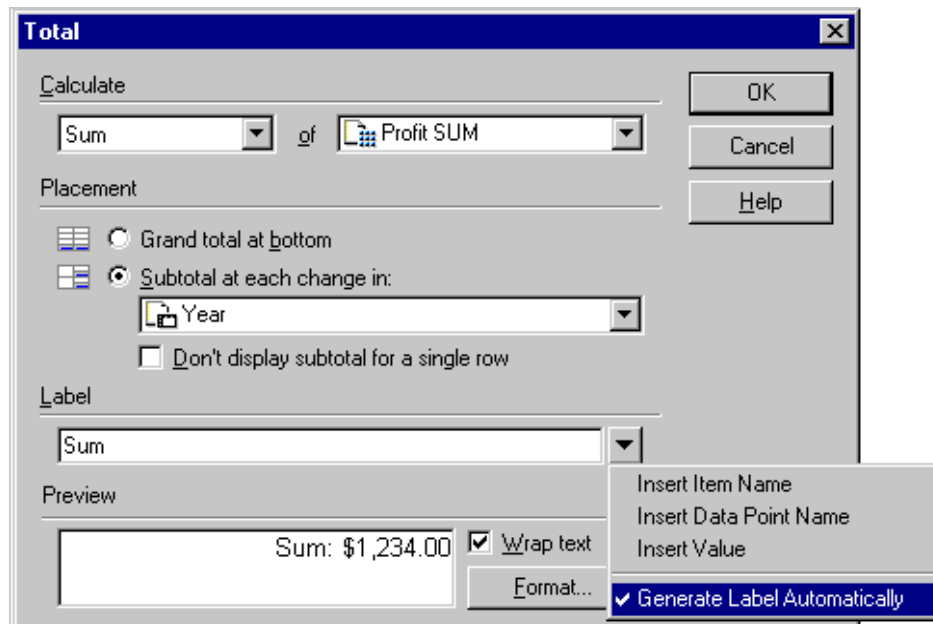
1. Select **Tools | Totals**. The Totals dialog box opens.



2. From the list View Totals for, select Profit SUM and click **New**. The Total dialog box opens.



3. Under Calculate, "Sum" of "Profit SUM" should appear in the two list boxes respectively
4. Under Placement, select "Subtotal at each change in", then choose Year from the pull down list below.

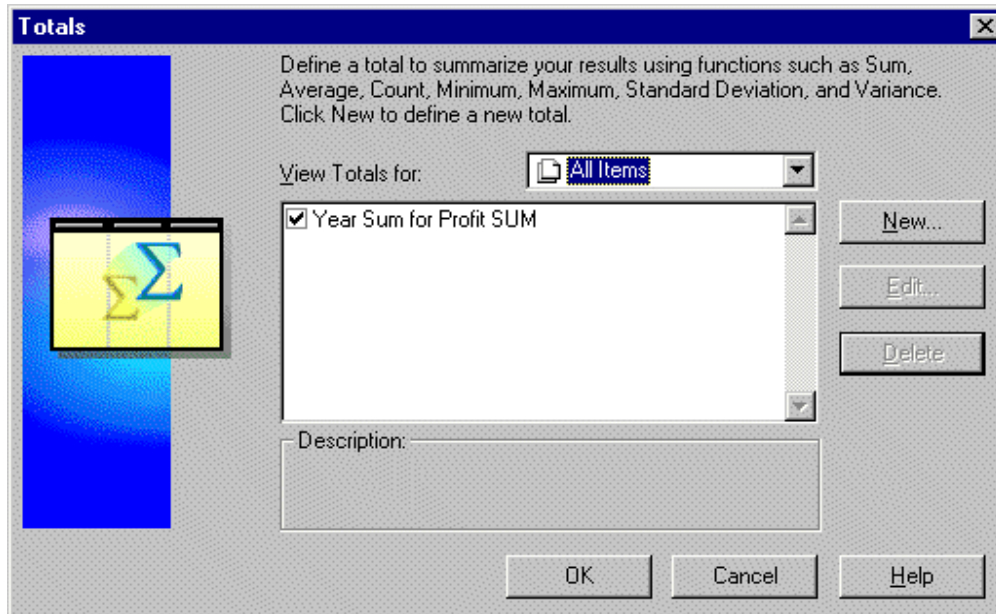


5. Preview the label for your calculation at the bottom of the dialog box.
6. Make sure that the Generate Label Automatically option is ticked.

Change the appearance of the calculation by clicking **Format**. The Format Data dialog box opens, allowing you to format the text and change the background color. Format your total for easy viewing in your Worksheet.

7. Click **OK** to confirm the total calculation.

In the Totals dialog box, notice that the total calculation you just created is selected.



8. Click **OK** to apply the selected total calculation.

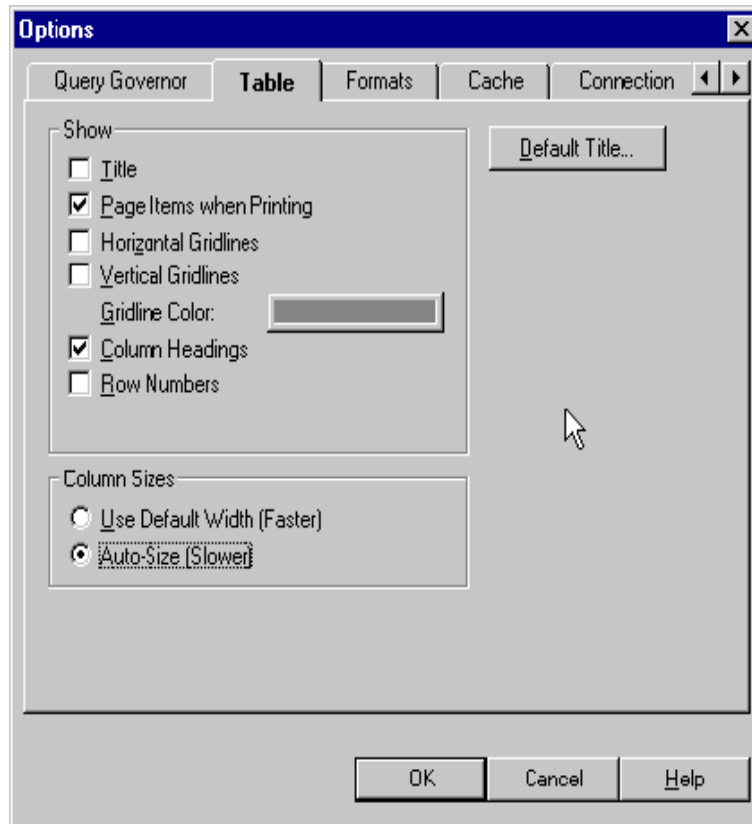
Page Items: Department: Video Rental ▼		
► Year	► Region	Profit SUM
1998	West	\$39,395.22
	Central	\$47,204.15
	East	\$71,766.31
		Sum: \$158,365.68
1999	West	\$37,777.44
	Central	\$43,812.58
	East	\$67,248.68
		Sum: \$148,838.70
2000	West	\$23,521.02
	Central	\$25,157.29
	East	\$40,401.56
		Sum: \$89,079.87

You now have a new row beneath each Year displaying the total profit for the Year.

1.14.3 To Autosize your Column Width

Discoverer enables you to Autosize the width of your columns so that all data can be viewed.

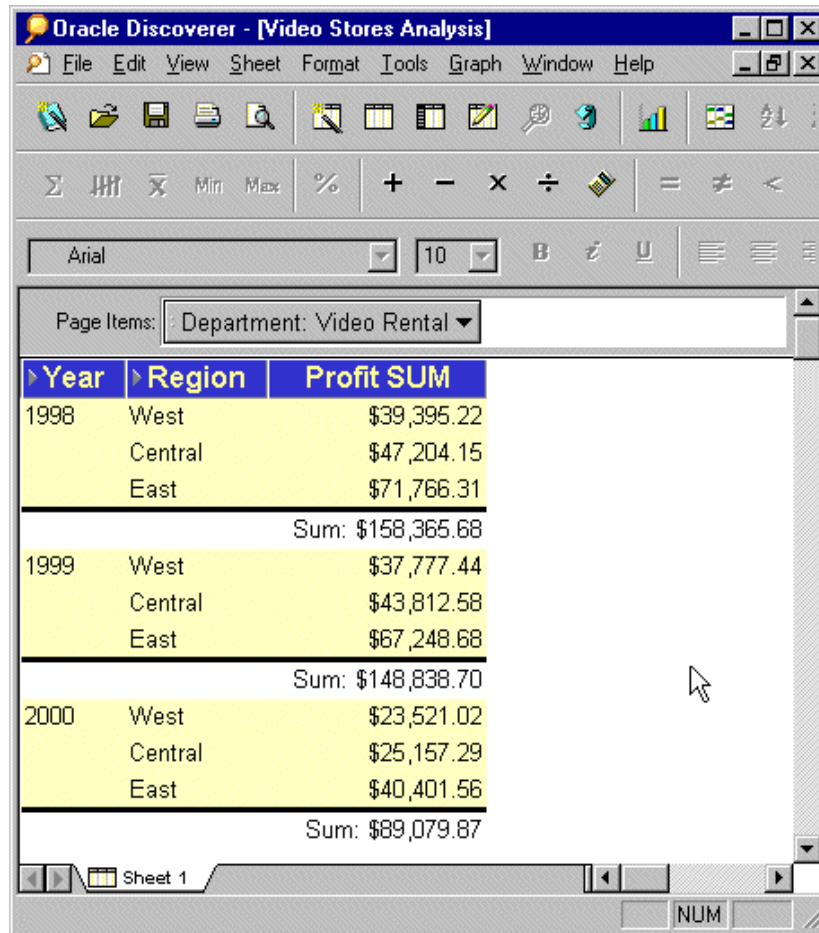
1. Select **Tools | Options**, and select the **Table** tab.



2. Under Column Sizes, select **Autosize**.
3. Click **OK**.

1.14.4 To Calculate Percentages

Next, you want to know the percentage that each Region's profits contribute against the company's total. You can easily add another calculation that calculates this percentage.



Oracle Discoverer - [Video Stores Analysis]

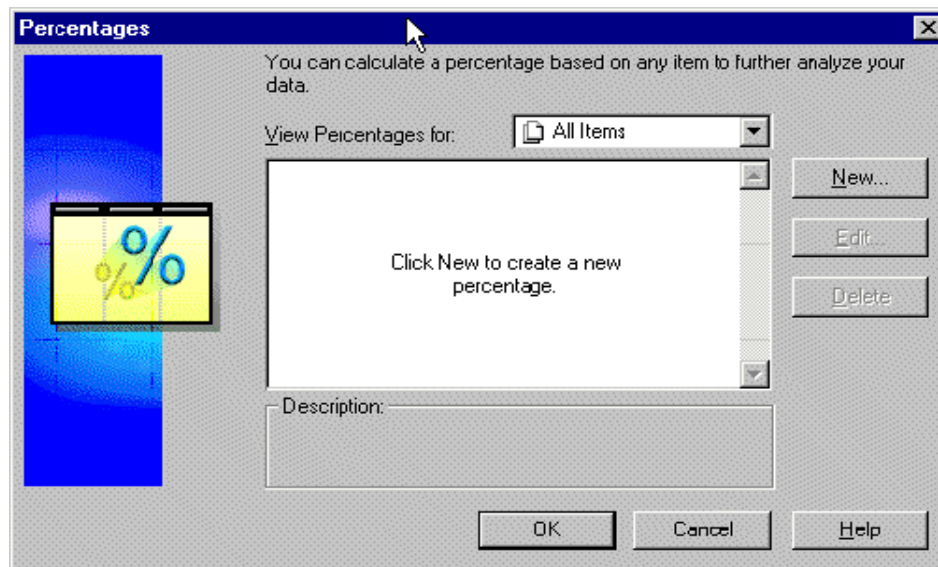
File Edit View Sheet Format Tools Graph Window Help

Page Items: Department: Video Rental

Year	Region	Profit SUM
1998	West	\$39,395.22
	Central	\$47,204.15
	East	\$71,766.31
		Sum: \$158,365.68
1999	West	\$37,777.44
	Central	\$43,812.58
	East	\$67,248.68
		Sum: \$148,838.70
2000	West	\$23,521.02
	Central	\$25,157.29
	East	\$40,401.56
		Sum: \$89,079.87

Sheet 1 NUM

1. Select **Tools | Percentages**. The Percentages dialog box opens.



2. Click **New**. The Percentage dialog box opens

3. Under **Calculate Percentages for**, select **Profit SUM**.
4. Under **Calculate as a Percentage of**, select **Each change in**, then choose **Year** from the pull down list.
5. Type '**% of Annual Profit**', in the **Column heading** box. Click on the format button to the right of this field to format the appearance of your percentage as you wish.
6. Click **OK** out of the Format Data dialog box to apply the specified formats and return to the Percentage dialog box.
7. Click **OK** to confirm your percentage calculation.
In the Percentages dialog box, notice that the percentage calculation you just created is selected.
8. Click **OK** to apply the selected percentage calculation.

Page Items: Department: Video Rental ▼			
Year	Region	Profit SUM	% of Annual Profit
1998	West	\$39,395.22	25%
	Central	\$47,204.15	30%
	East	\$71,766.31	45%
Sum: \$158,365.68			
1999	West	\$37,777.44	25%
	Central	\$43,812.58	29%
	East	\$67,248.68	45%
Sum: \$148,838.70			
2000	West	\$23,521.02	26%
	Central	\$25,157.29	28%
	East	\$40,401.56	45%
Sum: \$89,079.87			

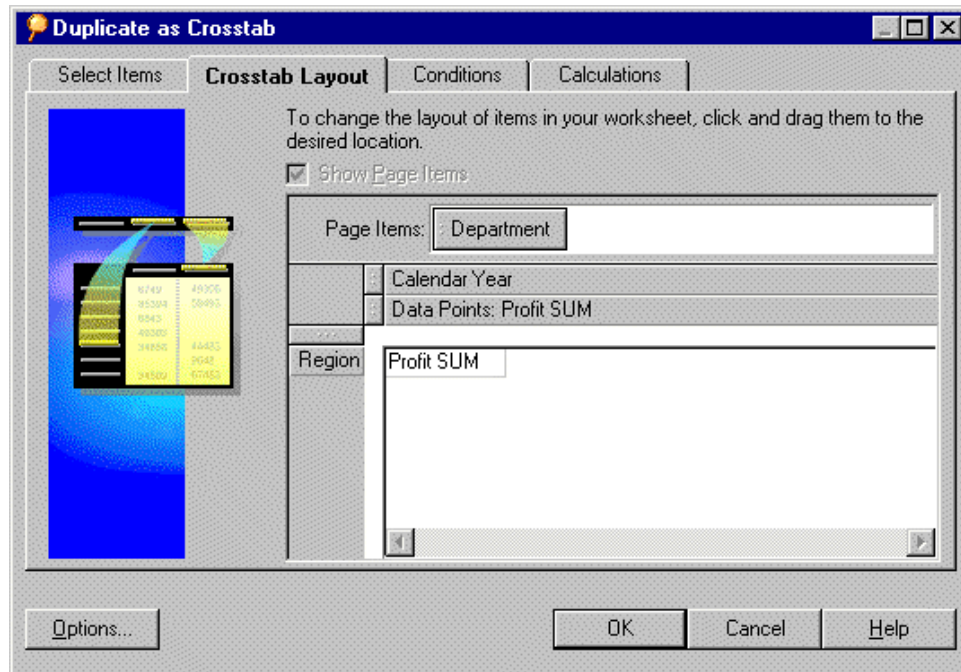
You have just calculated the Profit SUM as a percentage of total profit for each Year.

1.14.5 To Duplicate your Tabular Report to a Crosstab

Before you begin this section, rename Sheet 1 **Tabular Layout**. See "To rename a Worksheet" on page 1-36 for more information, and turn off the "% of Annual Profit" Percentage, (de-select the check box for this Item in the Percentages dialog).

1. Select **Sheet | Duplicate as Crosstab**. The Duplicate as Crosstab dialog box opens. Notice that the default page Item is Department, and the Region is placed on the left axis, with Calendar Year and Profit SUM on the top axis.

Of course you can rearrange the Crosstab layout to suit your needs, using the Discoverer skills that you have learned during this tutorial.



2. Click **OK**.

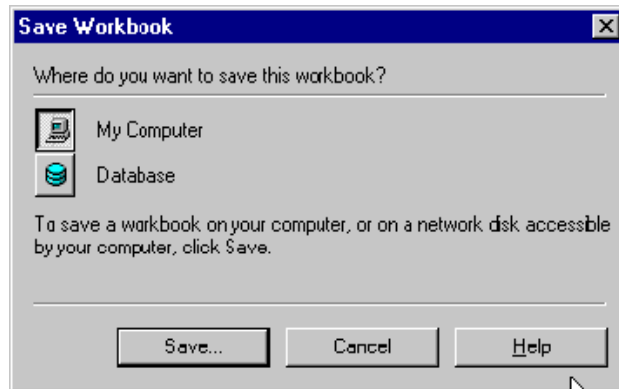
A new sheet is created with a crosstabular display of the same information. You can see that the total calculations that you created previously are recreated in this crosstab sheet.

Page Items: Department: Video Rental ▼				
	▶ 1998	▶ 1999	▶ 2000	
	Profit SUM	Profit SUM	Profit SUM	
▶ Central	\$47,204.15	\$43,812.58	\$25,157.29	
▶ East	\$71,766.31	\$67,248.68	\$40,401.56	
▶ West	\$39,395.22	\$37,777.44	\$23,521.02	

3. Rename this Worksheet **Crosstab Layout**.

1.14.6 To Save your New Workbook

1. Choose **File | Save**. The Save Workbook dialog box appears.



2. In the Save Workbook dialog box, choose **My Computer** to save this new Workbook to a directory on your computer or a network drive.
3. Click **Save**.
4. In the Save As dialog box, navigate to the [ORACLE_HOME]\discoverer\ directory and enter a file name.
5. Click **Save**.

You can also save your Workbook to the Discoverer database. Refer to section 1.7 for notes on saving your Workbooks.

1.14.7 To Add Items to a Worksheet

In the previous sections, you learned how to create your own Workbook. In this section, we return to the sample Discoverer Workbook, **Video Stores Analysis** to continue the tutorial.

At any time when you are working with a Discoverer Worksheet you can add or remove Items from the Worksheet. In this section, we are going to add the Cost Item to our Worksheet so that we can compare Costs with Profits.

1. In the **Video Stores Analysis** Workbook, click the Tabular Layout tab.

Oracle Discoverer - [Vidstr4.dis]

File Edit View Sheet Format Tools Graph Window Help

Analysis of Video Rentals and Video Sales

Page Items: Year: 1998

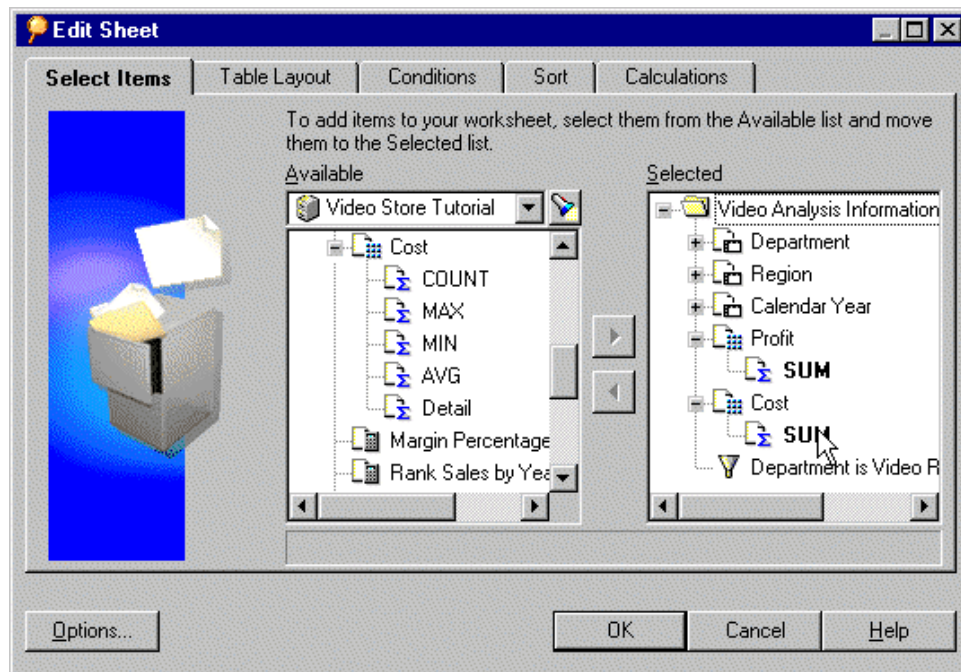
	Region	Department	Profit SUM
1	Central	Video Rental	\$47,204
2		Video Sale	\$67,084
3			Total for Central: \$114,288
4	East	Video Rental	\$71,766
5		Video Sale	\$108,558
6			Total for East: \$180,324
7	West	Video Rental	\$39,395
8		Video Sale	\$57,096
9			Total for West: \$96,491
10			Total for All Values: \$391,104

Tabular Layout Crosstab Layout

NUM

2. Select **Sheet | Edit Sheet**. The Edit Sheet dialog box opens.

Note: Those folders that are grayed out are inaccessible because those Items are not related, or joined, to the information already present in your Worksheet.



3. Open the **Video Analysis Information** folder by clicking on +, and drag the Item "Cost" to the Selected pane.

4. Click **OK** to add the Item to your Worksheet. You may have to wait for a few seconds for Discoverer to retrieve the data from the database.

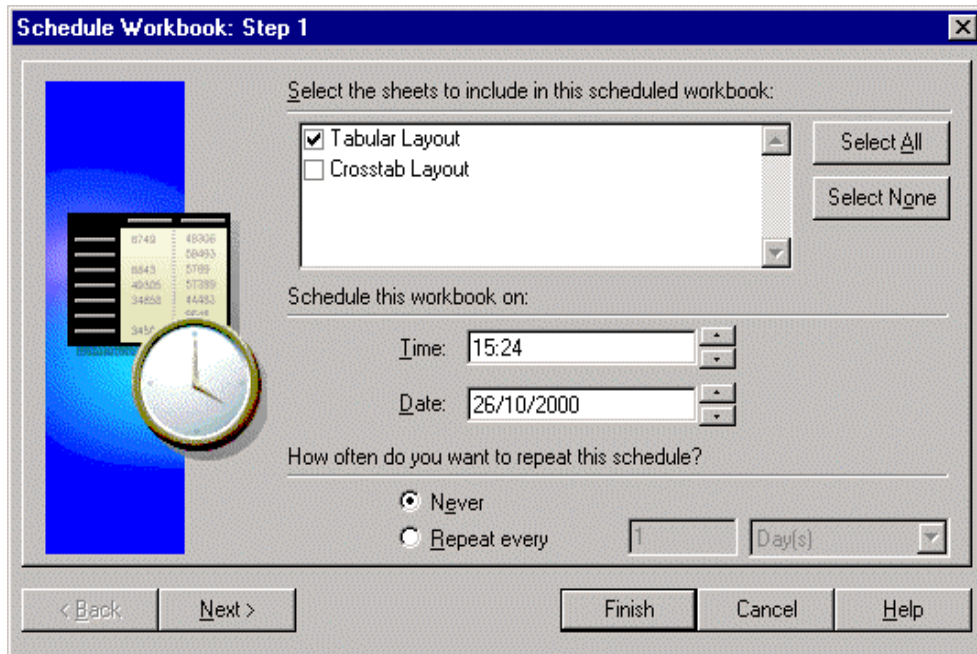
You can now use your Discoverer skills to analyze the Worksheet to see how Costs relate to Profits.

Page Items:		Year: 1998 ▼		
	Region	Department	Profit SUM	Cost SUM
1	Central	Video Rental	\$47,204	\$37,776.90
2		Video Sale	\$67,084	\$31,143.41
3	Total for Central		\$114,288	\$68,920.31
4	East	Video Rental	\$71,766	\$59,729.70
5		Video Sale	\$108,558	\$52,119.40
6	Total for East		\$180,324	\$111,849.10
7	West	Video Rental	\$39,395	\$25,951.34
8		Video Sale	\$57,096	\$22,621.86
9	Total for West		\$96,491	\$48,573.20
10	Total for All Values		\$391,104	\$229,342.61

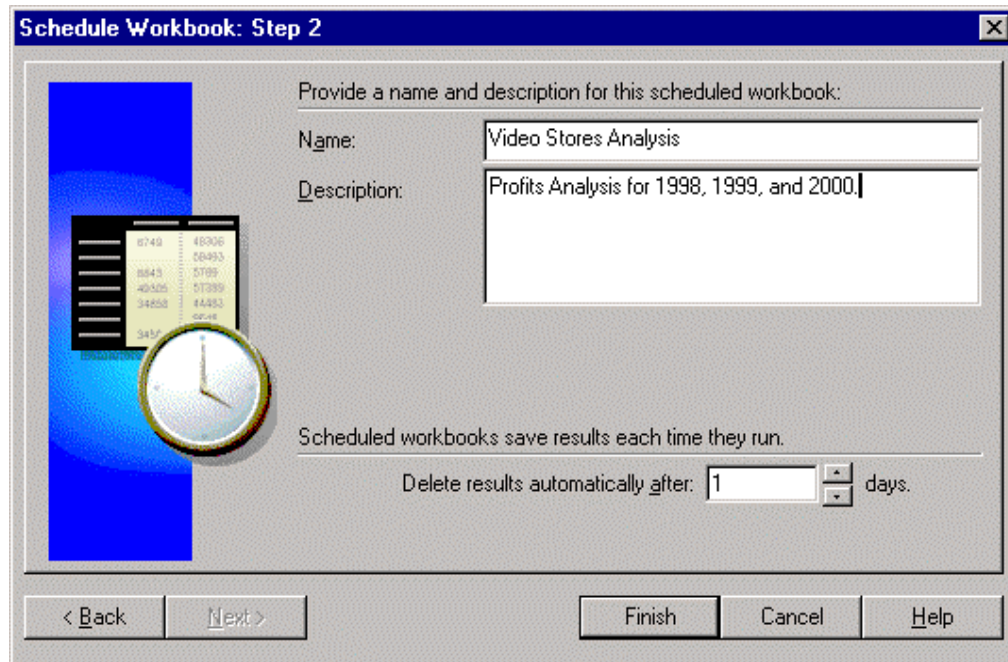
1.15 Scheduling a Workbook

It is likely that you will want to have your Workbook refreshed with the most up-to-date information at regular weekly or monthly intervals. This is especially helpful for large Workbooks that take a long time to run. Discoverer's **Schedule** facility enables you to schedule Workbooks to be updated at "off peak" times-even when you are not connected to Discoverer.

1. Open a Workbook, for example **Video Stores Analysis**, (Vidstr5.dis).
2. From the **File** menu, select **Schedule**. The Schedule Workbook Wizard opens. In our example, there are two Worksheets to choose from, Tabular Layout and Crosstab Layout.



3. In Step 1 of the Schedule Workbook wizard, select **Tabular Layout** from the list of sheets to choose from.
4. In the time field, schedule the sheet to run 5 minutes from now-leaving the date as today's date.
5. Under **How often do you want to repeat this schedule?**, Select **Never** so that the scheduled process does not reoccur.
6. Click **Next**. The **Schedule Workbook: Step 2** dialog box opens.



7. Name your Workbook with a meaningful name and type your own description.

8. Set your results to be deleted after **1** day.

9. Click **Finish**.

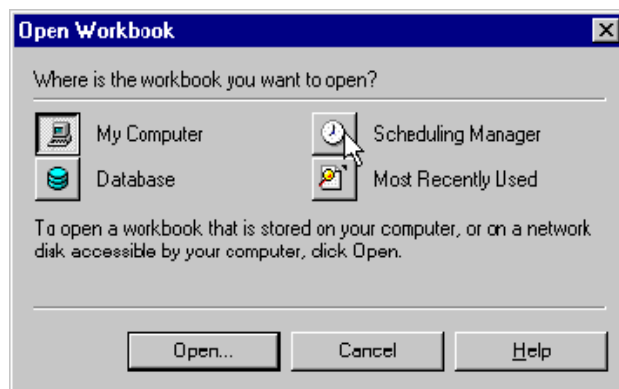
Depending on the size of the Workbook, the Scheduled Workbook will take a few minutes to create.

Your Workbook will be refreshed with the most up-to-date data in five minutes time. You can also set the Schedule to occur overnight so that you have up-to-date data when you begin work in the morning, or another suitable time.

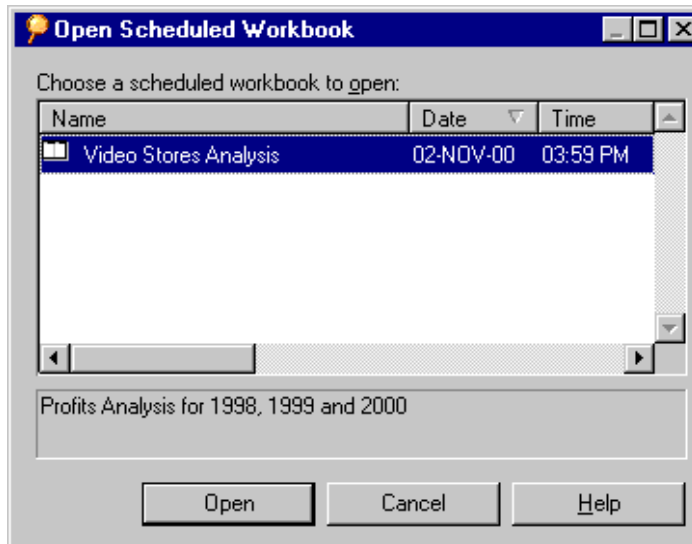
1.15.1 To Open your Scheduled Workbook

NOTE: Before starting this task, wait at least five minutes after completing the previous task.

1. Select **File | Open**. The Open Workbook dialog box opens.



2. Select **Scheduling Manager**, and then click **Open**. The **Open Scheduled Workbook** dialog box opens.



3. Select your previously scheduled Workbook, and click **Open** to display the Workbook.

4. Click **Yes** if you are asked if you want to run the query.

You see the results of your scheduled Workbook in a new Workbook which is identical to the Crosstab Layout sheet from your previously created Workbook.

End of the Discoverer Tutorial

Congratulations! You have just successfully learned the main features of Discoverer Desktop. You know how to pivot, drill, define various calculations, display results as graphs, schedule Workbooks, and more. You have all of the tools you need to begin producing meaningful reports for analysis, which ultimately means making better business decisions and increasing your company's productivity.